MECHANICAL DESIGN REFERENCE · EDITION 1.0 · NOVEMBER 2026

CNC Machine Tool Design Formula Handbook

A working calculation reference for mechanical design engineers of CNC machine tools — cutting mechanics, feed drives, spindles, structures, dynamics, guideways, thermal behaviour and precision metrology.

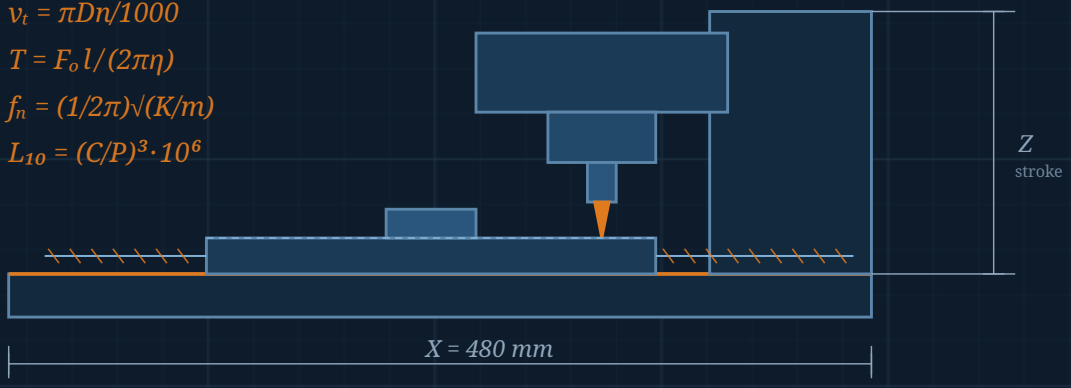
240	14	9	2
NUMBERED FORMULAS	DESIGN SECTIONS	DATA TABLES	WORKED EXAMPLES

$v_t = \pi D n / 1000$

$T = F_o l / (2\pi\eta)$

$f_n = (1/2\pi)\sqrt{(K/m)}$

$L_{10} = (C/P)^3 \cdot 10^6$



Prepared for: mechanical design & development of CNC turning, milling, grinding and special-purpose machine tools

Units: SI, with machining practice in mm · min · N · kW

How to use this handbook

Scope, notation conventions, unit system and the limits of every formula printed here.

Purpose and scope

This handbook collects the calculation set that a mechanical design engineer actually needs when specifying, sizing and checking the elements of a CNC machine tool. It runs from the process end (what the cut demands) through the drive train (what must deliver it) to the structure (what must resist it) and finally to the accuracy budget (what the customer measures). It is organised so that a designer can start in **Section 1** with a workpiece and a material, and finish in **Section 13** with a predicted positioning accuracy.

Every formula is numbered **F-001 … F-212**. Numbers are stable, so a calculation sheet, a design review note or a supplier query can cite them directly. Where a formula is empirical, or valid only inside a stated range, that is said explicitly in the note line under the entry — read the note before using the equation.

Notation conventions

Italic serif letters are variables (v_c, a_p). Roman letters are units, numbers or named constants. Subscripts identify the physical quantity: **c** cutting, **f** feed / friction, **p** passive / peak, **a** axial / acceleration, **r** radial / rear, **m** mean / motor, **n** nominal / normal, **0** reference or initial, **lim** limit, **all** allowable, **e** equivalent / endurance.

Angles are in degrees unless the argument is written in radians. Logarithms written **ln** are natural; **log** is base 10. **Δ** always means a change or an error, never an absolute value. Square brackets after an equation give the unit of the quantity on the left-hand side when it is not obvious from the variables.

Unit system

SI is used throughout, with the customary machine-tool working set: **mm** for length and diameter, **min** and **s** for time, **m/min** for cutting speed, **mm/min** for feed rate, **N** and **N·m** for force and torque, **kW** for power, **µm** for accuracy and deflection, **N/µm** for stiffness. Where a constant in an empirical formula carries units, the required input units are stated in the variable line — do not substitute SI values into those constants blindly. Pressure appears in **N/mm²** (= MPa) in strength work and in **bar** in hydraulics; 1 bar = 0.1 N/mm².

> **Three checks before any number leaves your desk.** (1) Dimensional consistency — if the constant is 9550 or 10^7, the input units are fixed and non-negotiable. (2) Validity range — empirical cutting-force, bearing-life and stability formulas are curve fits over a stated window. (3) Safety factor — machine tool elements are sized with the factors given in the note lines; a formula returns a value, not a decision.

Sources of the underlying relations: ISO 230 series (test code for machine tools), ISO 2892 and ISO 286 (tolerances and fits), ISO 3408 (ball screws), ISO 281 (rolling bearing life), ISO 6336 and the Lewis/AGMA gear methods, ISO 4413/4414 (fluid power), the Merchant chip-formation model, Taylor's tool-life law, and the standard catalogue methods of ball screw, linear guide, bearing and servo drive manufacturers. Empirical constants are reproduced as published; confirm against the current catalogue of the component you have actually selected.

Principal symbols

Symbol	Meaning	Unit	Symbol	Meaning
v_c	cutting speed	m/min	K	stiffness
n	rotational speed	rpm	J	mass moment of inertia
f, f_z	feed per revolution / per tooth	mm	ζ	damping ratio
v_f	feed rate (table speed)	mm/min	ω	angular velocity
a_p, a_e	depth of cut (axial) / width of cut (radial)	mm	α	acceleration; rake angle; thermal expansion coeff.
Q	material removal rate	cm³/min	μ	coefficient of friction
k_c	specific cutting force	N/mm²	η	efficiency
F_c, F_f, F_p	cutting, feed (thrust), passive force	N	σ, τ	normal stress, shear stress
P	power	kW	ε, δ	error / strain, deflection
T	torque; tool life (subscripted)	N·m; min	φ	geometric progression ratio
C, C_a	basic dynamic load rating	N	ρ	density; lead; radius of gyration
l, l_h	ball screw lead; lead in metres	mm; m	λ	lead angle; slenderness

The design sequence

The fourteen sections are ordered to follow a real machine-tool development project. Each stage consumes the output of the one before it, and the accuracy budget of the last stage is what the first nine exist to guarantee.

#	Stage	What is decided	Formulas
1	Process definition	Cutting speed, feed and depth of cut, converted into the force, torque and power that appear at the tool	F-001 – F-033
2	Productivity target	Cycle time, tool life, cost per piece and OEE — the commercial case the machine must satisfy	F-034 – F-049
3	Speed and feed drive	Speed range, number of steps, gear ratios and centre distances for stepped drives	F-050 – F-064
4	Feed axes	Ball screw, guideway, motor and drive: force, torque, inertia, life, stiffness, thermal growth	F-065 – F-105
5	Spindle	Power and torque envelope, bearing arrangement, tip stiffness, tool clamping interface	F-106 – F-128
6	Structure	Stiffness loop, deflection, joint compliance, then strength and fatigue of every load-carrying member	F-129 – F-170
7	Dynamics	Natural frequencies, isolation and chatter stability — the usable cutting window	F-171 – F-184
8	Guideways	Type selection, way pressure, stick-slip risk, hydrostatic and aerostatic sizing	F-185 – F-194
9	Thermal	Heat sources, temperature field, growth, compensation strategy and coolant sizing	F-195 – F-204
10	Auxiliary systems	Hydraulics and pneumatics for clamping, counterbalance, tool change and braking	F-205 – F-215
11	Verification	Accuracy budget, ISO 230-2 acceptance test, tolerance grades and fits	F-216 – F-240

Contents

Fourteen calculation sections, a worked axis design, data tables and a task index.

PROCESS — WHAT THE CUT DEMANDS

01	**Metal cutting mechanics & machining parameters**	F-001 – F-033	7
02	**Tool life, surface finish & process economics**	F-034 – F-049	11
03	**Spindle speed & feed drive design (stepped gearboxes)**	F-050 – F-064 + worked gearbox	13

DRIVE — WHAT MUST DELIVER IT

04	**Ball screws & linear drive elements**	F-065 – F-090	15
05	**Motion profiles, servo sizing & CNC kinematics**	F-091 – F-105	19
06	**Spindle system & rolling bearings**	F-106 – F-118	21
07	**Gears, belts & chains**	F-119 – F-128	23

STRUCTURE — WHAT MUST RESIST IT

08	**Structural stiffness, deflection & joints**	F-129 – F-142	26
09	**Strength, fatigue & machine elements**	F-143 – F-170	28
10	**Dynamics, vibration & chatter stability**	F-171 – F-184	32
11	**Guideways — sliding, rolling, hydrostatic & aerostatic**	F-185 – F-194	34

ENVIRONMENT & VERIFICATION

12	**Thermal behaviour & coolant systems**	F-195 – F-204 + heat-source table	36
13	**Hydraulics & pneumatics**	F-205 – F-215	38
14	**Precision, accuracy, tolerances & presses**	F-216 – F-240	40

APPLICATION

A	**Worked example — sizing a vertical machining centre X axis**	12 steps	43
B	**Design data tables**	9 tables	47
C	**Index of formulas**	F-001 – F-240	51
D	**Design release checklist**	34 checks	53

Quick-find by design task

Size a feed-axis servo motor	F-069 · F-084 – F-090 · F-093 · F-105 · F-106
Select and check a ball screw	F-065 – F-080
Select linear guideways	F-081 – F-083 · F-185 – F-194
Specify a spindle drive	F-011 – F-014 · F-105 · F-117 · F-118

Design a spindle bearing arrangement	F-106 – F-116
Estimate cutting forces and power	F-001 – F-015 · F-027 – F-031
Design a stepped spindle gearbox	F-050 – F-064 · F-119 – F-128
Check a machine structure for stiffness	F-129 – F-142 · F-147 – F-152
Prove out a shaft or weldment	F-143 – F-170
Predict vibration and chatter	F-171 – F-184
Control thermal error	F-078 · F-079 · F-195 – F-204
Build an accuracy budget / ISO 230 report	F-095 · F-096 · F-216 – F-232
Size hydraulics or pneumatics	F-205 – F-215
Quote cycle time and cost per part	F-040 – F-049
Design a stepped gearbox	F-050 – F-064 · proportioning rules table
Control heat and thermal error	F-195 – F-204 · heat-source table

Standards and test codes behind these formulas

Standard	Subject	Used in
ISO 1	Standard reference temperature for the specification and verification of geometrical quantities — 20 °C	F-203, F-222
ISO 230-1	Test code for machine tools: geometric accuracy under no-load or quasi-static conditions — the 21 geometric errors of a three-axis machine	F-220, F-221, F-229
ISO 230-2	Determination of accuracy and repeatability of positioning of numerically controlled axes	F-216 – F-219
ISO 230-3	Determination of thermal effects on the accuracy of a machine	Section 12
ISO 230-4	Circular tests for numerically controlled machine tools (ballbar method)	F-096, F-224
ISO 230-6	Determination of positioning accuracy on body and face diagonals — volumetric performance	F-229
ISO 230-7	Geometric accuracy of axes of rotation (rotary tables, spindles)	F-112 – F-114
ISO 286-1/-2	Code of limits and fits: tolerance system, tables of standard tolerances and limit deviations	F-225 – F-227, Table B8
ISO 2892	Derivation of the standard tolerance unit and the tolerance grades	F-225, F-226
ISO 281	Rolling bearings: dynamic load ratings and rating life	F-106 – F-108
ISO 3408	Ball screws: ratings, lead accuracy classes and mounting arrangements	F-065 – F-080
ISO 6336	Calculation of load capacity of spur and helical gears	F-119 – F-121
ISO 898-1	Mechanical properties of fasteners: bolts, screws and studs	F-156 – F-159
ISO 4413 / 4414	Hydraulic and pneumatic fluid power: general rules and safety requirements	Section 13
ISO 10791 series	Machining centres: acceptance conditions and test of performance	Appendix D checklist
ISO 13041 series	Turn-mill centres and turning centres: acceptance conditions	Appendix D checklist
ASME B5.54	Methods of performance evaluation for computer numerically controlled machining centres — volumetric performance	F-229
VDI/DGQ 3441	Determination and description of the positioning accuracy of machine axes — the basis of ISO 230-2	F-216 – F-219

SECTION 01

Metal cutting mechanics & machining parameters

F-001 – F-033
33 formulas

Start here. These formulas convert a workpiece material, a tool and a chosen productivity target into the speeds, feeds, forces, torques and powers that the rest of the machine must be designed to deliver. Everything downstream — motor rating, screw diameter, bearing preload, structural stiffness — is driven by the numbers produced in this section.

SPEEDS, FEEDS AND MATERIAL REMOVAL

F-001

Cutting speed (peripheral speed of the cut)

$$v_c = \frac{\pi D n}{1000} \quad [\text{m/min}]$$

v_c cutting speed (m/min) · D cutting diameter (mm) — workpiece diameter in turning, tool diameter in milling/drilling · n spindle speed (rpm)

F-002

Spindle speed required for a given cutting speed

$$n = \frac{1000\, v_c}{\pi D} \quad [\text{rpm}]$$

Rearrangement of F-001. In turning the diameter falls during the cut, so n must rise (CSS / constant-surface-speed mode) or the speed must be set at the largest diameter for a stepped-speed machine.

F-003

Feed rate from feed per revolution — turning, boring, drilling

$$v_f = n f \quad [\text{mm/min}]$$

f feed per revolution (mm/rev) · v_f axis feed rate (mm/min). This is the axis velocity the servo drive must command at full cutting load.

F-004

Feed rate from feed per tooth — milling

$$v_f = f_z z n \quad [\text{mm/min}]$$

f_z feed per tooth (mm) · z number of teeth · n cutter speed (rpm)

F-005

Material removal rate — turning

$$Q = a_p f v_c \quad [\text{cm}^3/\text{min}]$$

a_p depth of cut (mm) · f feed (mm/rev) · v_c cutting speed (m/min). The mixed units are deliberate and give cm³/min directly.

F-006

Material removal rate — milling

$$Q = a_e a_p v_f \quad [\text{mm}^3/\text{min}]$$

a_e radial width of cut (mm) · a_p axial depth of cut (mm) · v_f table feed (mm/min)

F-007

Material removal rate — drilling

$$Q = \frac{\pi D^2}{4} v_f \quad [\text{mm}^3/\text{min}]$$

Full-section removal. For core drilling substitute the annular area $\pi(D^2 - d^2)/4$.

CUTTING FORCES, POWER AND TORQUE

F-008

Specific cutting force (unit cutting force)

$$k_c = \frac{F_c}{a_p f} \quad [\text{N/mm}^2]$$

Force per unit undeformed chip area. Numerically equal to the specific cutting energy in J/mm³. Tabulated as $k_{c1.1}$ for a 1 mm chip thickness — see Table B1.

F-009

Main cutting force — turning

$$F_c = k_c a_p f \quad [\text{N}]$$

The tangential force that sets spindle torque, drive power, tool-holder clamping and the torsional load on the drive train.

F-010

Correction of k_c for chip thickness

$$k_c = k_{c,ref} \left(\frac{f_{ref}}{f} \right)^{m_c}$$

f_{ref} reference chip thickness, usually 0.3 or 1.0 mm as tabulated · m_c material exponent, 0.15–0.35 (Table B1). Specific force **rises** as the chip gets thinner — do not use a coarse-feed k_c for finishing passes.

F-011

Cutting power

$$P_c = \frac{k_c Q}{60 \times 10^3} = \frac{F_c v_c}{60 \times 10^3} \quad \text{[kW]}$$

First form: Q in cm³/min, k_c in N/mm². Second form: F_c in N, v_c in m/min. This is power **at the cutting edge**, before transmission losses.

F-012

Motor power required

$$P_m = \frac{P_c}{\eta} \quad \text{[kW]}$$

η total transmission efficiency = product of gearbox, belt, bearing and spindle efficiencies. Typical machine-tool values: gear drive 0.90–0.95 per stage, belt 0.95–0.98, integral spindle 0.85–0.92. Size the motor on the **worst simultaneous** cut, not the average.

F-013

Torque from power and speed

$$T = 9550 \frac{P}{n} \quad \text{[N·m]}$$

P in kW, n in rpm. The single most-used conversion in machine design; 9550 = 60·1000/(2π).

F-014

Spindle torque from the cutting force

$$T = \frac{F_c D}{2000} \quad \text{[N·m]}$$

D cutting diameter in mm. Cross-check against F-013 — the two must agree within the transmission efficiency.

F-015

Resultant cutting force

$$F_R = \sqrt{F_c^2 + F_f^2 + F_p^2} \quad \text{[N]}$$

F_c tangential (cutting) · F_f axial (feed) · F_p radial (passive/back). Rule of thumb for turning with 90° approach: $F_f \approx 0.3–0.4\ F_c$, $F_p \approx 0.3–0.5\ F_c$. F_p does no work but deflects the workpiece, the tool and the structure — it governs accuracy, not power.

CHIP FORMATION (MERCHANT MODEL)

F-016

Chip thickness ratio

$$r = \frac{h_1}{h_2} = \frac{\sin \varphi}{\cos(\varphi - \alpha)}$$

h_1 undeformed chip thickness · h_2 deformed chip thickness · φ shear angle · α rake angle (positive by convention). Measuring h_2 in a test gives φ without a dynamometer.

F-017

Shear angle from the chip ratio

$$\tan \varphi = \frac{r \cos \alpha}{1 - r \sin \alpha}$$

A larger φ means a shorter, thicker shear plane, lower force and lower power — the physical reason a positive rake and a sharp edge reduce spindle load.

F-018

Merchant's minimum-energy relation

$$2\varphi + \beta - \alpha = 90° \quad \rightarrow \varphi = 45° + \frac{\alpha}{2} - \frac{\beta}{2}$$

β friction angle at the rake face. An idealisation — real materials deviate by 5–15° — but it correctly predicts that increasing rake or reducing rake-face friction raises the shear angle.

F-019

Shear strain in the primary deformation zone

$$\gamma_s = \cot \varphi + \tan(\varphi - \alpha)$$

Typical values 2–5. Drives the heat generated per unit volume and therefore the fraction of power that must be removed by coolant (F-195).

F-020

Shear and normal force on the shear plane

$$F_s = F_c \cos\varphi - F_f \sin\varphi \;;\quad F_n = F_c \sin\varphi + F_f \cos\varphi$$

Resolves measured dynamometer forces onto the shear plane. F_s/A_s gives the shear flow stress of the work material at cutting strain and strain rate.

F-021

Friction angle and coefficient at the rake face

$$\beta = \alpha + \arctan\frac{F_f}{F_c} \;;\quad \mu = \tan\beta$$

Useful for judging whether a coating or a coolant is actually reducing rake-face friction — compare β before and after at identical v_c, f, a_p.

MILLING GEOMETRY

F-022

Cutter engagement (contact) angle

$$\theta_{en} = \arccos\left(1 - \frac{2\,a_e}{D}\right) \quad \text{[deg]}$$

a_e radial depth of cut · D cutter diameter. Slotting gives θ_{en} = 180°; half-slotting 90°; light skin milling a few degrees. Governs the force pulsation frequency that excites the structure.

F-023

Average chip thickness — milling

$$h_m = f_z \sqrt{\frac{a_e}{D}} \cdot \frac{180}{\theta_{en}} \quad \text{[mm]}$$

Use h_m — not f_z — as the chip thickness when entering k_c charts. In light-engagement high-speed milling h_m is far below f_z, so the feed per tooth must be raised to keep the edge in its working window.

F-024

Effective number of teeth in cut

$$z_e = z\,\frac{\theta_{en}}{360}$$

Sets the mean cutting force and the torque ripple. With $z_e < 1$ the process is intermittent and the spindle sees full force pulsation at tooth-passing frequency (F-183).

F-025

Approach distance — face milling

$$l_c = \frac{1}{2}\left(D - \sqrt{D^2 - 4a_e{}^2}\right) \quad \text{[mm]}$$

Distance the table travels from first tooth contact to full radial engagement. Needed for the true cutting time (F-040) and for the length over which the drive must already be at full feed — a direct input to the acceleration profile (F-092).

DRILLING, GRINDING AND THREAD CUTTING

F-026

Drill point approach length

$$l_e = \frac{D}{2\tan\kappa_r} \approx 0.3\,D \quad \text{[mm]}$$

κ_r half point angle. For the standard 118° point, tan 59° = 1.66, so $l_e \approx 0.3D$. Add it to the drill travel — omitting it underestimates hole-cycle time by 10–30% on short holes.

F-027

Drilling torque and thrust (empirical)

$$T = C_T\,D^{1.8}\,f^{0.8}\,K_m \;;\quad F_f = C_F\,D\,f^{0.75}\,K_m$$

D drill diameter (mm), f feed (mm/rev), C_T, C_F material constants from machining-data handbooks, K_m correction for hardness, point angle and coolant. **Empirical** — good to ±25%; measure on the actual material for a critical spindle or feed-axis rating.

F-028

Drilling power

$$P = \frac{2\pi\,n\,T}{60 \times 10^3} \quad \text{[kW]}$$

T in N·m, n in rpm. The Z-axis (spindle head) feed drive must additionally hold F_f against the ball screw and the guide preload.

F-029

Grinding — wheel speed and speed ratio

$$v_s = \frac{\pi D_s n_s}{1000} \;;\quad q = \frac{v_s}{v_w}$$

D_s wheel diameter (mm) · n_s wheel speed (rpm) · v_w workpiece speed (m/min) · q speed ratio, typically 60–150 for surface grinding, 100–300 for creep feed. Wheel peripheral speed is limited by the bond: vitrified ≈ 35–60 m/s, resinoid ≈ 60–100 m/s.

F-030

Grinding — specific removal rate and specific energy

$$Q' = a_e v_w \;\; [\text{mm}^2/\text{s}] \;;\quad u = \frac{P}{Q} \;\; [\text{J/mm}^3]$$

Q' is removal rate per unit wheel width, with v_w in mm/s. Grinding specific energy is 10–60 J/mm³ for steel — an order of magnitude above turning — which is why grinding spindles are power-rich and coolant-rich relative to their removal rate.

F-031

Grinding ratio (G-ratio)

$$G = \frac{\text{volume of workpiece removed}}{\text{volume of wheel worn}}$$

Sets dressing interval and wheel cost per part. G falls sharply with wheel speed above the bond limit and with insufficient coolant — the two parameters a grinding-machine designer controls least well.

F-032

Thread-cutting gear train ratio

$$i = \frac{\text{pitch to be cut}}{\text{leadscrew pitch}} = \frac{z_{\text{driver}}}{z_{\text{driven}}}$$

The fundamental relation of a thread-cutting gear box. On a CNC machine the same relation is enforced electronically by the spindle-encoder-to-axis gearing (electronic gear box ratio = lead per spindle revolution ÷ axis lead).

F-033

Power-screw torque — taps, lead screws, clamping screws

$$T = \frac{F d_m}{2} \cdot \frac{\tan\lambda + \mu \sec\beta_n}{1 - \mu \tan\lambda \sec\beta_n} \;\; [\text{N·m}]$$

F axial load (N) · d_m pitch diameter (m) · λ = arctan($l/\pi d_m$) lead angle · β_n half thread profile angle (30° for ISO metric, 29° for Acme) · μ friction coefficient. For a tapping cycle apply with $\mu \approx 0.15$ and the tap pitch diameter; for a sliding (Acme/trapezoidal) lead screw this replaces F-065 with $\eta \approx 0.3$–0.5.

SECTION 02

Tool life, surface finish & process economics

F-034 – F-049
16 formulas

A machine tool is bought to make parts at a cost. These formulas convert the cutting parameters of Section 1 into tool life, achievable surface finish, cycle time and cost per piece — the numbers that justify a spindle rating, a tool magazine size or an automatic pallet changer in the design specification.

TOOL LIFE

F-034

Taylor's tool life equation

$$v_c \, T^n = C$$

T tool life to a stated wear criterion, usually VB = 0.3 mm (min) · n Taylor exponent · C speed at T = 1 min. Typical n: HSS 0.10–0.15, carbide 0.20–0.30, ceramic 0.40–0.60, CBN/PcD 0.25–0.40. Log-log plot is a straight line of slope $-1/n$.

F-035

Extended tool life equation

$$v_c \, T^n f^{n_f} a_p^{\,n_p} = C$$

Exponents rank the influence of the parameters: for carbide turning typically $n \approx 0.25$, $n_f \approx 0.4$–0.7, $n_p \approx 0.15$–0.3. Feed hurts tool life more than depth of cut — the basis of the "high-feed, moderate-depth" roughing strategy that a machine must be stiff and powerful enough to support.

F-036

Tool life at a changed cutting speed

$$T_2 = T_1 \left(\frac{v_1}{v_2} \right)^{1/n}$$

With n = 0.25 a 20% speed increase cuts tool life to 41% of its former value. This is the arithmetic that decides whether a "faster spindle" is a productivity gain or a consumables problem.

SURFACE FINISH

F-037

Theoretical peak roughness — nose radius

$$R_t = \frac{f^2}{8\, r_\varepsilon} \; ; \quad R_a \approx \frac{f^2}{32\, r_\varepsilon} \qquad \text{[mm]}$$

r_ε tool nose radius (mm). Halving the feed quarters the roughness; doubling the nose radius halves it. Real surfaces are 2–5× worse because of built-up edge, vibration, feed marks of the drive and coolant-induced edge chipping — treat F-037 as a floor, not a prediction.

F-038

Theoretical peak roughness — sharp tool with approach angles

$$R_t = \frac{f}{\tan \kappa_r + \cot \kappa_r'} \qquad \text{[mm]}$$

κ_r principal (lead) approach angle · κ_r' auxiliary approach angle. For a 90°/45° tool the denominator is 2, so $R_t = f/2$.

F-039

Vibration-induced roughness contribution

$$R_{a,\text{total}} \approx \sqrt{R_{a,\text{ideal}}^{\,2} + A_{\text{vib}}^{\,2}}$$

A_{vib} peak-to-mean relative vibration amplitude between tool and workpiece (mm), measured or taken from the frequency response at the tooth-passing frequency. Links Section 10 dynamics directly to a surface-finish specification.

TIME AND COST

F-040

Basic cutting time — turning

$$t_m = \frac{L}{n\, f} = \frac{\pi\, D\, L}{1000\, v_c\, f} \qquad \text{[min]}$$

$L = l_w + l_{in} + l_{out}$: workpiece length plus tool approach and overtravel. For a 90° tool $l_{in} \approx a_p/\tan \kappa_r + 2$ mm, l_{out} = 2–5 mm.

F-041

Basic cutting time — milling

$$t_m = \frac{L}{v_f}, \quad L = l_w + l_c + l_o \quad \text{[min]}$$

l_c approach from F-025 · l_o overtravel 2–5 mm. On a machining centre the same expression gives the rapid-to-cut transition length that the acceleration profile must absorb.

F-042

Basic cutting time — drilling

$$t_m = \frac{l_w + l_e + l_o}{v_f} \quad \text{[min]}$$

l_e from F-026. For deep holes add the peck-retract strokes: $t = (n_{peck} + 1)\cdot t_{per\ peck} + n_{peck}\cdot t_{retract}$.

F-043

Total cycle time per piece

$$T_c = t_m + t_h + \frac{t_m}{T}\, t_{ct} \quad \text{[min]}$$

t_h handling, load/unload, index and non-cutting time (min) · T tool life (min) · t_{ct} time per tool change (min). The third term is the tool-change time amortised over the piece. Non-cutting time typically dominates on machining centres — which is why rapid traverse, acceleration and tool-change time are design parameters, not luxuries.

F-044

Tool cost per piece

$$C_{tool} = \frac{t_m}{T}\,(C_e + t_{ct}\, k_m)$$

C_e cost of one cutting edge (insert + share of holder) · k_m machine and labour cost per minute. Multiply by the number of tools cutting the piece.

F-045

Cost per piece

$$C_{piece} = k_m\, T_c + C_{tool} + C_{mat} + C_{fix}$$

C_{fix} amortised fixture, programming and set-up share. This is the figure a machine's specification sheet is really selling against.

F-046

Economic tool life (minimum cost per piece)

$$T_o = \left(\frac{1}{n} - 1\right) \frac{C_e}{k_m} \quad \text{[min]}$$

Differentiating F-045 with respect to speed. With n = 0.25 the multiplier is 3 — the economic tool life is three times the edge cost divided by the machine-minute cost. Expensive machines (high k_m) should run faster and change tools more often.

F-047

Tool life for maximum production rate

$$T_p = \left(\frac{1}{n} - 1\right) t_{ct} \quad \text{[min]}$$

Always shorter than T_o. If the machine has a fast tool changer ($t_{ct} \approx$ 1–2 s chip-to-chip) the maximum-rate strategy is close to the minimum-cost one; with a slow changer they diverge sharply. A direct argument for investing in tool-change speed.

F-048

Optimum cutting speed

$$v_o = \frac{C}{T_o^{\,n}} \quad \text{[m/min]}$$

From F-034 with $T = T_o$ (cost) or T_p (rate). Check that the resulting speed is inside the machine's constant-power range (F-105) and below the tool-holder speed rating.

F-049

Overall equipment effectiveness

$$\mathrm{OEE} = A \times P \times Q \quad \text{[\%]}$$

A availability = run time ÷ planned time · P performance = ideal cycle time ÷ actual cycle time · Q quality = good parts ÷ total parts. Machine design influences A (reliability, MTBF, maintainability) and P (acceleration, rapid traverse, tool-change time, stiffness-limited feeds).

SECTION 03

Spindle speed & feed drive design (stepped gearboxes)

F-050 – F-064
15 formulas

The classical machine-tool design problem: given a speed range, produce a geometric progression of spindle speeds from a minimum number of gears. Still directly relevant for heavy-duty machines, gear-head lathes, gearboxes behind a two-range motor, and the feed boxes of thread-cutting machines.

SPEED STEPPING

F-050

Speed range of the spindle

$$R_n = \frac{n_{max}}{n_{min}}$$

Set from the smallest and largest diameter to be machined at the economic cutting speeds of the tooling. Typical CNC turning centres R_n = 100–200; milling machines 50–150; a wide-range AC motor alone gives 100–300 in the constant-power region.

F-051

Geometric progression (structural) ratio

$$\varphi = R_n^{1/(z-1)} = \sqrt[z-1]{R_n}$$

z number of speed steps. Only a geometric progression gives a **constant** relative loss of economic cutting speed across the whole range — this is why it, and not an arithmetic progression, is the machine-tool standard.

F-052

Value of the *i*-th speed step

$$n_i = n_{min}\,\varphi^{i-1}\,, \quad i = 1 \ldots z$$

Round each result to the nearest preferred number (F-055) and check that the actual φ between adjacent rounded values stays within ±5%.

F-053

Number of speed steps required

$$z = 1 + \frac{\ln R_n}{\ln \varphi}$$

Round up to an integer, then prefer a z that factorises into 2s and 3s (e.g. 6, 8, 12, 18) so the box can be built as a cascade of two- and three-speed groups.

F-054

Loss of economic cutting speed

$$A = \frac{\varphi - 1}{\varphi} \times 100 \quad [\%]$$

The worst-case percentage by which the available speed falls below the ideal. φ = 1.26 → 21% loss; φ = 1.12 → 11%; φ = 1.06 → 6%. A stepless (inverter/AC) drive makes $A \to 0$, which is the whole commercial argument for it.

F-055

Preferred-number (Renard) series derivation

$$\varphi_{R40} = 10^{1/40} = 1.059\,; \quad \varphi_{R20} = 10^{1/20} = 1.122\,; \quad \varphi_{R10} = 10^{1/10} = 1.259$$

Standard machine-tool φ values are 1.06, 1.12, 1.26, 1.41, 1.58, 2.00 — respectively the 40th, 20th, 10th, 5th, 2.5th and 1st roots of 10. Speeds derived from them are the ISO/R20 preferred speed series (Table B3).

GEAR TRAIN GEOMETRY

F-056

Transmission ratio of a gear pair

$$i = \frac{n_{driven}}{n_{driver}} = \frac{z_{driver}}{z_{driven}} = \frac{d_{driver}}{d_{driven}}$$

Machine-tool convention: $i < 1$ is a reduction (speed falls, torque rises). Torque on the driven shaft = $T_{driver}/i \cdot \eta$.

F-057

Overall ratio of a gear train

$$i_{tot} = i_1\, i_2 \ldots i_k = \frac{\text{product of driver teeth}}{\text{product of driven teeth}}$$

Idler gears change direction only and do not appear in the ratio. Total efficiency $\eta_{tot} = \eta^k$ for k meshing pairs.

F-058

Structural formula and group transmission range

$$z = p_1(X_1) \cdot p_2(X_2) \cdot p_3(X_3) \ldots , \quad X_i = \varphi^{\text{group range}}$$

p_i number of speed steps produced by group i; X_i the number of speed steps by which the two speeds of one gear pair in that group are separated. Example: z = 18 = 3(1)·3(3)·2(9). The last group always has the largest X; keep $X_{max} \le p_{max} \cdot p_{next}$ so the ray diagram does not cross.

F-059

Practical ratio limits in a stepped gearbox

$$\frac{1}{4} \le i \le 2 \quad \text{(sliding gear clusters)} \; ; \quad \frac{1}{8} \le i \le 4 \quad \text{(fixed trains with clutches)}$$

Beyond these limits the gears become too unequal in size for one centre distance, and sliding clusters jam. Split a large ratio into two stages.

F-060

Centre distance and module

$$a = \frac{m(z_1 + z_2)}{2} \; ; \quad a_{\text{helical}} = \frac{m_n(z_1 + z_2)}{2 \cos \beta} \quad \text{[mm]}$$

m transverse module · m_n normal module · β helix angle. All gear pairs in one shaft group must share the same centre distance so that sliding clusters stay in mesh.

F-061

Constant sum of teeth within a group

$$z_1 + z_2 = \frac{2a}{m} = \Sigma z = \text{constant}$$

For every gear pair on the same two shafts. Given Σz and a required ratio i, the pinion has $z_1 = \Sigma z/(1 + 1/i)$ teeth — round to an integer and accept the small resulting speed error, or correct it in an adjacent group.

F-062

Minimum number of teeth (no profile shift)

$$z_{\min} = \frac{2}{\sin^2 \alpha} = 17 \quad \text{for } \alpha = 20°$$

α pressure angle. Below this the tooth undercuts and loses strength; use profile shift ($x > 0$) to run fewer teeth, at the cost of a changed operating centre distance.

F-063

Feed range and feed progression ratio

$$R_f = \frac{f_{\max}}{f_{\min}} \; ; \quad \varphi_f = R_f^{1/(z_f - 1)}$$

Identical logic to the spindle box. On CNC machines feeds are stepless, so the design task shifts to guaranteeing **torque at minimum feed** — the axis must hold full cutting force at 1 mm/min without stalling or cogging.

F-064

Pitch diameter and gear peripheral speed

$$d = m\,z \; ; \quad v = \frac{\pi\, d\, n}{60 \times 10^3} \quad \text{[m/s]}$$

Peripheral speed sets the gear accuracy grade required: v < 2 m/s grade 9–10 acceptable; 2–6 m/s grade 8; 6–15 m/s grade 7; > 15 m/s grade 6 or better, ground. Machine-tool gearboxes are usually grade 6–7 and run in oil.

SECTION 04

Ball screws & linear drive elements

F-065 – F-090
26 formulas

The feed axis is where a CNC machine most often fails its specification: too soft, too hot, resonant at rapid traverse, or with a life shorter than the warranty. This section covers the complete ball-screw and linear-guide selection chain — force, torque, speed, buckling, life, stiffness, thermal growth and inertia.

TORQUE AND AXIAL FORCE

F-065

Ball screw drive torque

$$T_1 = \frac{F_a\, l}{2\pi\, \eta} \quad \text{[N·m]}$$

F_a axial force on the nut (N) · l lead (m) · η ball screw efficiency, 0.90–0.95. Add bearing and seal friction (typically +5–10%) to obtain the torque the motor must deliver.

F-066

Back-drive (holding) torque

$$T_2 = \frac{F_a\, l\,(2 - 1/\eta)}{2\pi} \quad \text{[N·m]}$$

Torque the axial load exerts back on the screw. At η = 0.95 this is 0.95·$F_a l/2\pi$ — nearly the full drive torque, so a high-efficiency ball screw **will** back-drive. Vertical axes therefore need a brake, a counterbalance or a self-locking screw (F-067 with η < 0.5).

F-067

Screw efficiency from the lead angle

$$\eta = \frac{\tan\lambda}{\tan(\lambda + \rho')}\,, \quad \lambda = \arctan\frac{l}{\pi\, d_m}\,, \quad \rho' = \arctan\frac{\mu}{\cos\beta}$$

Applies to sliding (trapezoidal/Acme) screws, where $\mu \approx 0.10$–0.15 gives $\eta \approx 0.30$–0.50 and the screw is self-locking. For recirculating-ball screws use the measured catalogue efficiency instead — rolling contact makes the sliding model invalid.

F-068

Axial force the drive must produce

$$F_a = F_{cut} + \mu\, W + m\, a \pm m\, g \sin\theta \quad \text{[N]}$$

F_{cut} feed-direction cutting force (N) · μ guideway friction (0.002–0.005 rolling, 0.05–0.15 sliding with way film) · W weight normal to the guideway (N) · m moving mass (kg) · a axis acceleration (m/s²) · θ axis inclination from horizontal. Take the sign that maximises F_a; for a vertical axis mg is fully additive on the up-stroke.

SPEED LIMITS

F-069

Critical (whirling) speed of the screw shaft

$$n_{cr} = f\,\frac{d_r}{L^2} \times 10^7 \quad \text{[rpm]}$$

d_r screw root (root-of-thread) diameter (mm) · L unsupported distance between bearing supports (mm) · f end-fixity factor: **3.9** fixed–free, **11.5** supported–supported, **17.7** fixed–supported, **25.5** fixed–fixed. Equivalent first-principles form: $n_{cr} = (60/2\pi)\cdot(\lambda L)^2\cdot\sqrt{(EI/\rho A)}/L^2$ with $(\lambda L)^2$ = 3.52 / 9.87 / 15.4 / 22.4.

> **Note:** critical speed scales with d_r and with **1/L²** — halving the unsupported span quadruples it. On long-stroke machines the screw, not the motor, sets the rapid-traverse speed. Moving the nut changes L continuously, so check the worst position (nut nearest mid-span).

F-070

Permissible rotational speed and *DN* limit

$$n_{perm} = 0.8\, n_{cr}\,; \quad n\, d_m \le (DN)_{lim}$$

The 0.8 factor is the standard catalogue safety margin on critical speed. The DN value (bore-equivalent diameter in mm × rpm) limits ball centrifugal loading and recirculation impact; typical limits 70 000–150 000 for standard nuts, higher for high-speed designs. Both limits apply — use the lower.

BUCKLING AND LIFE

F-071

Buckling load of the screw shaft

$$P_{cr} = \frac{\pi^2 E I}{(K L)^2} \quad , \quad I = \frac{\pi d_r^4}{64} \quad [N]$$

Effective-length factor K: 0.50 fixed–fixed, 0.70 fixed–pinned, 1.00 pinned–pinned, 2.00 fixed–free. E = 207 GPa for bearing steel. Governs compressive loading on long vertical and high-thrust axes.

F-072

Safety factor against buckling

$$n_s = \frac{P_{cr}}{F_{a,max}} \geq 2 \ldots 4$$

Use 4 for vertical axes and for any axis where the load can shock (crash, clamping failure). If n_s is marginal, increase the root diameter rather than shortening the stroke.

F-073

Ball screw rating life

$$L = \left(\frac{C_a}{F_a}\right)^3 \times 10^6 \quad [\text{revolutions}]$$

C_a basic dynamic axial load rating (N) · F_a equivalent axial load (N). The cube law means a 26% load increase halves the life. L is the L_{10} life — 90% of a population survives it.

F-074

Rating life in operating hours

$$L_h = \frac{10^6}{60\, n_m} \left(\frac{C_a}{F_m}\right)^3 \quad [h]$$

n_m mean rotational speed (rpm) · F_m mean axial load from F-075. Design targets: 20 000 h for general machining centres, 30 000–40 000 h for high-duty or unmanned-operation machines.

F-075

Mean (equivalent) axial load over a duty cycle

$$F_m = \sqrt[3]{\frac{\Sigma F_i^3 \, n_i \, t_i}{\Sigma \, n_i \, t_i}} \quad [N]$$

Cubic mean, because life depends on the cube of the load. Build the duty cycle from the actual part programme: rapid traverse (low force, high speed), cutting feed (high force, low speed), dwell, return. Using the arithmetic mean understates damage badly when short high-load segments exist.

STIFFNESS, THERMAL AND PRELOAD

F-076

Axial stiffness of the screw system

$$K_s = \frac{A E}{L} = \frac{\pi d_r^2 E}{4 L} \; ; \quad \frac{1}{K_{axis}} = \frac{1}{K_s} + \frac{1}{K_{nut}} + \frac{1}{K_{bear}} + \frac{1}{K_{mount}} \quad [N/mm]$$

L is the distance from the nut to the **fixed** bearing support, so shaft stiffness varies with table position — the axis is softest at mid-span. K_{nut} and K_{bear} come from the catalogue; a preload-ed nut is roughly 2× stiffer than an unpregloaded one. Divide by 1000 for N/µm.

F-077

Positioning error caused by drive compliance

$$\varepsilon = \frac{F_a}{K_{axis}} \quad [mm]$$

A load-dependent, direction-dependent error that appears as a bidirectional repeatability band in an ISO 230-2 test (F-216). Example: 8 kN cutting force on a 120 N/µm axis gives 67 µm of lost motion — unacceptable, and the reason high-thrust axes are pre-tensioned and double-nutted.

F-078

Thermal elongation of the screw

$$\Delta L = \alpha \, L \, \Delta T \quad [mm]$$

α = 11.7 × 10^{-6} /K for steel. A 1 m screw warming 5 K grows 5.9 µm — comparable to the whole positioning tolerance of a precision machine. This is the dominant thermal error source on long-stroke axes and the reason for pre-tensioning, screw cooling and thermal compensation.

F-079

Pre-tension force for thermal compensation

$$F_{\text{pre}} = E\,A\,\alpha\,\Delta T_{\text{target}} = E\,\frac{\pi\,d_{\text{r}}^{\,2}}{4}\,\alpha\,\Delta T_{\text{target}} \quad \text{[N]}$$

The screw is stretched cold by the amount it would grow when hot, so the operating length stays near nominal. ΔT_{target} is the expected steady-state rise (typically 2–4 K). The pre-tension also raises axial stiffness by roughly 2×. Check that F_{pre} plus cutting force stays inside the buckling and bearing limits.

F-080

Heat generated by screw friction

$$P_{\text{loss}} = T_1\,\omega\,\left(\frac{1}{\eta} - 1\right) = \frac{F_{\text{a}}\,l\,\omega}{2\pi}\,\left(\frac{1}{\eta} - 1\right) \quad \text{[W]}$$

ω screw angular velocity (rad/s). At η = 0.92 and 3000 rpm under 10 kN this is ≈ 500 W dumped into the screw — enough to raise an uncooled 1 m screw by 8–12 K. Feed it into F-195 to size screw cooling.

F-081

Linear guideway rating life

$$L = 50\,\left(\frac{f_{\text{H}}\,C}{f_{\text{w}}\,P}\right)^{3} \quad \text{[km]}$$

C basic dynamic load rating (N) · P equivalent load per block (N) · f_{H} hardness factor (1.0 at ≥ 58 HRC, falling to ≈ 0.7 at 50 HRC) · f_{w} load factor 1.0–1.5 (smooth) up to 2.0–3.5 (shock/vibration). Convert to hours with the mean axis speed.

F-082

Static safety factor of a guideway block

$$f_{\text{s}} = \frac{C_0}{P_0} \geq 1.5 \ldots 4$$

C_0 basic static rating · P_0 maximum static equivalent load including shock. Below about 1.5 the raceway takes permanent brinelling marks that destroy smooth motion and accuracy — the usual cause of "the machine was crashed once and never ran true again".

F-083

Equivalent block load with an overturning moment

$$P = f_{\text{w}}\,\left(F + K_{\text{M}}\,\frac{M}{D}\right) \quad \text{[N]}$$

M overturning moment about the block centre (N·mm) · D and K_{M} model-specific constants published in the guide manufacturer's table. Always resolve the cutting force into the three moment axes (M_{A}, M_{B}, M_{C}) and take the worst block of the set — the corner block, never the average.

INERTIA AND TORQUE DEMAND

F-084

Acceleration torque

$$T_{\text{a}} = J\,\frac{\Delta\omega}{\Delta t} = \frac{J\,\Delta n}{9.55\,t_{\text{a}}} \quad \text{[N·m]}$$

J total inertia referred to the motor shaft (kg·m²) · Δn speed change (rpm) · t_{a} acceleration time (s). On modern machining centres T_{a} is 2–5× the cutting torque — acceleration, not cutting, sizes the feed motor.

F-085

Total and RMS motor torque

$$T_{\text{m}} = T_{\text{a}} + T_{\text{load}}\;;\quad T_{\text{rms}} = \sqrt{\frac{\Sigma\,T_i^{\,2}\,t_i}{\Sigma\,t_i}} \quad \text{[N·m]}$$

Select the motor so that T_{peak} ≤ maximum (intermittent) torque and T_{rms} ≤ rated (continuous) torque. Include the deceleration segment — its magnitude equals the acceleration torque but with regenerated energy (F-104).

F-086

Inertia of a solid cylinder (shaft, screw, coupling)

$$J = \frac{m\,d^2}{8} = \frac{\pi\,\rho\,L\,d^4}{32} \quad \text{[kg·m²]}$$

d outside diameter (m) · L length (m) · ρ density (kg/m³). Note the fourth-power dependence on diameter: a screw one size larger costs far more inertia than it buys in stiffness.

F-087

Inertia of a translating mass referred to the screw

$$J_{\text{lin}} = m\,\left(\frac{l}{2\pi}\right)^2 \quad \text{[kg·m²]}$$

l screw lead (m). A 500 kg table on a 10 mm lead reflects 12.7 × 10⁻⁴ kg·m² — usually larger than the screw's own inertia, and the term that decides whether the motor can meet the acceleration target.

F-088

Inertia and torque through a reduction

$$J_{\text{ref}} = \frac{J_{\text{load}}}{i^2} \quad ; \quad T_{\text{ref}} = \frac{T_{\text{load}}}{i\,\eta}$$

i gear or belt ratio (> 1 for reduction). A reduction of 2 cuts reflected inertia fourfold but doubles required motor speed — the classic stiffness-versus-speed trade of a belt-driven screw.

F-089

Inertia ratio check

$$\frac{J_{\text{load}}}{J_{\text{motor}}} \le R_{\text{J,max}}$$ (5 … 10 high dynamics, 10 … 30 standard, up to 50 with a stiff coupling)

Exceeding the drive manufacturer's limit degrades settling time, excites the screw's torsional mode and can make the position loop unstable regardless of gain. If the ratio is too high, reduce it with a gear/belt stage or a larger motor — not by detuning the servo.

F-090

Rack-and-pinion and belt drive equivalence

$$J_{\text{ref}} = m\,r^2 \quad ; \quad F = \frac{T\,\eta}{r} \quad ; \quad v = \omega\,r$$

r pinion pitch radius (m). Used on gantry and long-stroke machines where a ball screw would be too slender. Note the absence of the $1/L^2$ penalty — but also the lower stiffness ($K = \eta \cdot T/r \cdot \ldots$ limited by tooth and bearing compliance) and the backlash that must be taken out by a split pinion or dual drive with preload.

SECTION 05

Motion profiles, servo sizing & CNC kinematics

F-091 – F-105
15 formulas

How the machine actually moves: the velocity profile the control generates, the acceleration and jerk the structure must survive, the following error the contour pays for, and the interpolation geometry behind every block of G-code.

MOTION PROFILES

F-091

Trapezoidal profile — acceleration time and distance

$$t_a = \frac{v_{max}}{a} \;;\; s_a = \frac{v_{max}^2}{2a}$$

v_{max} rapid traverse (m/s) · a acceleration (m/s²). At 60 m/min and 1 g the ramp takes 0.10 s and covers 51 mm — longer than many short moves, which is why small-hole drilling cycles never reach programmed rapid.

F-092

Total positioning time (trapezoidal, no jerk limit)

$$T = \frac{L}{v_{max}} + \frac{v_{max}}{a} \quad [s]$$

Valid when $L \geq v^2_{max}/a$. For shorter moves the profile is triangular and $T = 2\sqrt{(L/a)}$. This is the number that goes into the tool-change and pallet-change cycle-time budget.

F-093

Acceleration achievable within the torque limit

$$a = \frac{\frac{T_m \, i \, \eta \, 2\pi}{l} - F_f}{m} \quad [m/s^2]$$

T_m available motor torque (N·m) · i reduction · l lead (m) · F_f friction and gravity load (N) · m moving mass (kg). Compare with the acceleration the structure and the workpiece can tolerate — the lower of the two is the specification.

F-094

Jerk-limited (S-curve) profile

$$J = \frac{a_{max}}{t_j} \;;\; v_{max} < \frac{a^2}{J} \Rightarrow \text{triangular (no constant-acceleration phase)}$$

t_j jerk ramp time (s). S-curve profiles trade a slightly longer move for a large reduction in structural excitation and marking. Typical machine-tool jerk limits: 5–50 m/s³ for milling centres, up to 200 m/s³ for high-speed die-sinking machines with light spindles.

FOLLOWING ERROR AND CONTOURING

F-095

Following (servo lag) error

$$e = \frac{v_f}{K_v} \quad [mm]$$

v_f commanded axis velocity (mm/s) · K_v position-loop gain (s⁻¹). A steady-state lag on **every** axis during motion. At 10 m/min and K_v = 10 s⁻¹ the lag is 16.7 µm per axis — visible on a diagonal contour as a corner round-off.

F-096

Circularity error on a circular interpolation

$$\Delta R \approx \frac{v_f^2}{2 \, R \, K_v^2} \;;\; \Delta R_{mismatch} = v_f \left| \frac{1}{K_{vx}} - \frac{1}{K_{vy}} \right|$$

First term: radius error with matched gains (a small circle, second-order). Second term: the ellipse produced by unequal axis gains — the signature seen on a ballbar circularity test. Equalising K_v across axes is the cheapest accuracy improvement available on any CNC machine.

F-097

Resultant feed rate of simultaneous axes

$$F = \sqrt{\Sigma \, v_i^2} \quad [mm/min]$$

The programmed path feed is the vector sum. Each axis therefore runs at $v_i = F \cdot \cos \theta_i$ — check every axis against its own velocity and torque limit, not just the path.

F-098

Block execution time

$$t = \frac{\sqrt{\Sigma\, \Delta x_i^{\,2}}}{v_f} \quad \text{[min]}$$

The basis of look-ahead processing: if t is shorter than the control's block-processing time or the deceleration time at a corner, the machine becomes **block-limited** and never reaches the programmed feed. Diagnose with the axis-load meter, not the programme.

INTERPOLATION GEOMETRY

F-099

Circular interpolation — centre offsets and radius

$$I = x_c - x_s \;;\quad J = y_c - y_s \;;\quad R = \sqrt{I^2 + J^2}$$

Subscripts s = start, c = centre. In incremental (I/J) mode the offsets are from the start point; in R mode the control solves the two-circle intersection and picks the short or long arc by the sign of R.

F-100

Arc length and sweep time

$$s = R\,\theta \quad \text{[mm, } \theta \text{ in rad]} \;;\quad t = \frac{s}{v_f}$$

The angular velocity of the arc is $\omega = v_f / R$ — on a small radius at high feed this becomes the limiting factor for the rotary axis in cylindrical interpolation.

F-101

Helical interpolation

$$\text{lead} = \frac{v_{f,\text{axial}}}{n_{\text{rot}}} \;;\quad v_{\text{path}} = \sqrt{v_{f,\text{circ}}^{\,2} + v_{f,\text{axial}}^{\,2}}$$

Used for helical milling of large bores and for thread milling. The three axes must be gain-matched or the helix pitch errors by the differential following error of F-095.

F-102

Cutter radius compensation offset path

$$x' = x \pm r_t \cos\theta \;;\quad y' = y \pm r_t \sin\theta$$

θ tangent angle of the programmed path · r_t cutter radius · + for left (G41), − for right (G42) of the direction of travel. At an inside corner where the offset path self-intersects the control must insert a gouge-avoidance move — the source of most cutter-comp alarms.

F-103

Polar / cylindrical coordinate transform

$$r = \sqrt{x^2 + y^2} \;;\quad \theta = \arctan\frac{y}{x} \;;\quad x_{\text{cyl}} = \theta\, r_c \quad \text{[mm, unwrapped]}$$

For cylindrical interpolation (X–C) and polar interpolation (X–Y from C). Near $r \to 0$ the unwrapped linear speed of the C axis tends to zero while its angular speed tends to infinity — a hard limit the control must clamp.

DRIVE ENERGY

F-104

Regenerative braking energy and resistor rating

$$E = \frac{1}{2}\, J\,(\omega_1^{\,2} - \omega_2^{\,2}) \;;\quad P_R = \frac{E \times (\text{number of decelerations per cycle})}{t_{\text{cycle}}} \quad \text{[W]}$$

Energy returned to the DC bus on every deceleration. If it exceeds the drive's internal absorption capacity the bus over-voltage trips the drive — the classic "machine faults at rapid traverse" symptom. Size the external braking resistor on the **worst** axis combination decelerating simultaneously.

F-105

Motor torque–speed characteristic (two regions)

$$T = T_{\text{rated}} \quad \text{for } n \le n_b \;;\quad T = \frac{9550\, P_{\text{rated}}}{n} \quad \text{for } n > n_b$$

n_b base (rated) speed. Below it the drive is torque-limited (constant torque); above it field-weakened and power-limited, so torque falls as $1/n$. Every spindle and feed specification must state which region the duty point lies in — a spindle rated 15 kW at 6000 rpm may deliver only 5 kW at 2000 rpm.

SECTION 06

Spindle system & rolling bearings

F-106 – F-118
13 formulas

The spindle is the one element whose behaviour the customer feels directly in every part. These formulas cover bearing life and speed limits, the stiffness seen at the tool tip, thermal growth, and the tool-clamping interface.

BEARING LIFE AND SPEED

F-106

Basic rating life of a rolling bearing

$$L_{10} = \left(\frac{C}{P}\right)^{p} \times 10^{6} \quad \text{[revolutions]}$$

C basic dynamic load rating (N) · P equivalent dynamic load (N) · p = 3 for ball bearings, 10/3 for roller bearings. L_{10} = life reached or exceeded by 90% of a population. For a machine-tool spindle the target is normally 20 000–30 000 h at the duty-cycle mean speed.

F-107

Rating life in operating hours

$$L_{h} = \frac{10^{6}}{60\,n}\left(\frac{C}{P}\right)^{p} \quad \text{[h]}$$

For a variable-speed spindle use the cubic (ball) or 3.33-power (roller) mean of the speed–load pairs, exactly as in F-075.

F-108

Equivalent dynamic bearing load

$$P = X F_{r} + Y F_{a} \quad \text{[N]}$$

F_r, F_a radial and axial loads · X, Y bearing-type factors from the catalogue, dependent on F_a/C_0 and on F_a/F_r versus e. For angular-contact bearings in a spindle arrangement the induced axial force from the radial load must be included — a paired (back-to-back or face-to-face) set carries an internal thrust even with no external axial load.

F-109

Bearing speed limit — *DN* value

$$DN = d\,(\text{mm}) \times n\,(\text{rpm}) \le (DN)_{\text{lim}}$$

A proxy for ball centrifugal force and cage speed. Typical grease-lubricated precision angular-contact limits: 0.9–1.5 × 10^6; oil-air or oil-mist 1.5–2.5 × 10^6. Above roughly 1.5 × 10^6 use ceramic (Si_3N_4) balls — see F-110 for why.

F-110

Centrifugal force on a bearing ball

$$F_{c} = 2.26 \times 10^{-11}\, D_{b}^{3}\, d_{m}\, n^{2} \quad \text{[N]}$$

D_b ball diameter (mm) · d_m bearing pitch diameter (mm) · n inner-ring speed (rpm). Grows with the **cube** of ball diameter, which is why high-speed spindles use many small balls, and with the density — silicon-nitride balls are 40% lighter, cutting F_c by the same proportion and letting the preload stay effective at speed.

F-111

Bearing friction torque and heat generation

$$M_{f} = \frac{1}{2}\,\mu\, F\, d_{m}\;;\quad P_{\text{loss}} = M_{f}\,\omega = 1.047 \times 10^{-7}\,\mu\, F\, d_{m}\, n \quad \text{[W]}$$

$\mu \approx 0.0010$–0.0015 for precision angular-contact ball bearings, 0.0015–0.0025 for cylindrical rollers · F total load including preload (N) · d_m pitch diameter (mm) · n rpm. Preload raises F and therefore the heat — the central trade-off of spindle design between stiffness and thermal growth. Feed this number into F-195 to predict the temperature field.

SPINDLE STIFFNESS AND THERMAL BEHAVIOUR

F-112

Stiffness at the tool tip — rigid shaft on elastic bearings

$$\frac{1}{K_{\text{tip}}} = \left(\frac{l+a}{l}\right)^{2}\frac{1}{K_{f}} + \left(\frac{a}{l}\right)^{2}\frac{1}{K_{r}} \quad \text{[N/µm]}$$

a tool-tip overhang from the front bearing (mm) · l bearing span (mm) · K_f, K_r radial stiffness of front and rear bearing sets. The lever arms are squared: doubling the overhang makes the spindle roughly four times softer. This is the strongest single argument for a short tool and a long bearing span.

F-113

Tool-tip deflection — elastic shaft, rigid bearings

$$\delta = \frac{F\,a^{2}(l+a)}{3\,E\,I} \quad \text{[mm]}$$

Add to F-112 in series for the true tip compliance. With a hollow shaft $I = \pi(D^4 - d^4)/64$ — hollowing a spindle nose costs very little stiffness for a large saving in mass and therefore in acceleration torque.

F-114

Spindle thermal growth

$$\Delta L = \alpha\, L\, \Delta T\ ; \quad \Delta D = \alpha\, D\, \Delta T \quad \text{[mm]}$$

α = 11.7 × 10^{-6}/K steel. A 300 mm spindle nose-to-gauge-line distance warming 10 K grows 35 µm in Z — larger than the entire positioning tolerance of a precision machine. Control it by cooling the bearing outer rings, matching the housing material to the shaft, or measuring and compensating.

TOOL INTERFACE

F-115

Torque capacity of a taper tool connection

$$T_{\text{cap}} = \frac{\mu\, F_{\text{clamp}}\, R_{\text{m}}}{\cos \gamma} \quad \text{[N·m]}$$

F_{clamp} drawbar clamping force (N) · R_{m} mean contact radius of the taper (m) · γ taper half-angle (8.25° for 7:24 BT/CAT, 8°17′50″ for HSK) · μ ≈ 0.08–0.15 dry, lower when lubricated. If the cutting torque exceeds T_{cap} the tool slips in the taper — the classic cause of a spun holder and a scored nose.

F-116

Required drawbar clamping force

$$F_{\text{clamp}} \geq S\,(F_{\text{axial}} + F_{\text{centrifugal}})\ , \quad S = 1.5 \ldots 2.5$$

F_{axial} axial cutting force pulling the tool out (N) · $F_{\text{centrifugal}}$ mass-force pull-out of the holder at speed. On a 7:24 taper the cone also opens under centrifugal load above ≈ 12 000 rpm, losing contact stiffness — the reason HSK and similar hollow-shank, face-contact tapers are specified for high speed.

F-117

Power delivered at the tool

$$P_{\text{tool}} = P_{\text{motor}}\, \eta_{\text{spindle}} = \frac{T_{\text{tool}}\, n}{9550} \quad \text{[kW]}$$

Quote the machine's power at the **spindle nose across the whole speed range**, not the motor nameplate. A belt-driven 15 kW motor through a 2:1 step-up delivers 15 kW only above the base speed of the reduced side.

F-118

Spindle acceleration time

$$t_{\text{a}} = \frac{J_{\text{sp}}\, \Delta\omega}{T_{\text{m}} - T_{\text{load}}} = \frac{J_{\text{sp}}\, \Delta n}{9.55\,(T_{\text{m}} - T_{\text{load}})} \quad \text{[s]}$$

J_{sp} total rotating inertia of motor rotor, coupling, gears, spindle shaft, chuck and tool (kg·m²). Include the workpiece in turning — a 200 kg chuck-and-part can triple J_{sp} and is the usual reason a lathe misses its quoted ramp time.

SECTION 07

Gears, belts & chains

F-119 – F-128
10 formulas

Power transmission between motor and spindle or motor and screw. Strength checks for the gearbox, and the tension and efficiency relations for belt and chain drives.

F-119

Gear tooth bending stress (Lewis equation)

$$\sigma_b = \frac{W_t}{b\, m\, Y\, K_v} \le \sigma_{all} \quad [\text{N/mm}^2]$$

W_t tangential tooth load (N) · b face width (mm) · m module (mm) · Y Lewis form factor (function of tooth count and pressure angle; ≈ 0.32 for 20 teeth, 0.40 for 30, 0.48 for 50 at 20°) · K_v velocity factor. Lewis treats the tooth as a static cantilever loaded at the tip — conservative for the tip-loaded case, unconservative if the load is shared; use ISO 6336 for a rated design.

F-120

Dynamic (velocity) factor — AGMA

$$K_v = \left(\frac{A}{A + v_t}\right)^B, \quad A = 50 + 56(1 - B), \quad B = \frac{(12 - Q_v)^{2/3}}{4}$$

v_t pitch-line velocity (m/s) · Q_v AGMA/ISO transmission-accuracy number (6 for a precision machine-tool gear, 8–11 for commercial gearing). Higher accuracy and lower speed both push K_v toward 1.

F-121

Gear contact (pitting) stress — ISO 6336 form

$$\sigma_H = Z_E\, Z_H\, Z_\varepsilon \sqrt{\frac{F_t\, K_A\, K_v\, K_{H\beta}\, K_{H\alpha}}{d_1\, b} \cdot \frac{u + 1}{u}} \quad [\text{N/mm}^2]$$

Z_E elasticity factor (≈ 190 √(N/mm²) steel/steel) · Z_H zone factor (≈ 2.5 at 20°) · Z_ε contact-ratio factor · d_1 pinion pitch diameter · $u = z_2/z_1$ gear ratio · K_A application, $K_{H\beta}$ face-load-distribution, $K_{H\alpha}$ transverse-load factors. Contact stress, not bending, usually governs a hard-finished machine-tool gearbox.

F-122

Involute gear geometry

$$d = m\, z; \quad p = \pi\, m; \quad d_b = d \cos\alpha; \quad d_a = d + 2m(1 + x); \quad d_f = d - 2.5m$$

α pressure angle (20° standard) · x profile-shift coefficient · subscripts b, a, f = base, tip (addendum), root (dedendum) circles. Addendum = m, dedendum = 1.25m, giving 0.25m of bottom clearance.

F-123

Contact ratio

$$\varepsilon_\alpha = \frac{\sqrt{r_{a1}^2 - r_{b1}^2} + \sqrt{r_{a2}^2 - r_{b2}^2} - a \sin\alpha_w}{\pi\, m \cos\alpha} \ge 1.2$$

Mean number of tooth pairs in contact. Below 1.2 the load transfers abruptly between pairs and the gearbox becomes noisy and dynamically loaded — a common defect in a hastily proportioned spindle two-speed gear.

F-124

Gear mesh efficiency

$$\eta = 1 - \pi\, \mu \left(\frac{1}{z_1} + \frac{1}{z_2}\right)$$

μ ≈ 0.05–0.10 for lubricated steel gears. Gives 0.985–0.995 per mesh — a two-stage gearbox loses 2–3%, which must appear in F-012.

F-125

Belt tension ratio (Euler–Eytelwein)

$$\frac{T_1}{T_2} = e^{\mu\,\theta}; \quad T_1 - T_2 = F_t \quad [\text{N}]$$

T_1 tight side, T_2 slack side · μ belt/pulley friction (0.3–0.5 for V-belts expressed as an equivalent, 0.25–0.35 flat) · θ angle of wrap on the **smaller** pulley (rad) · F_t effective tangential force. For a V-belt use $\mu' = \mu/\sin(\gamma/2)$, γ = groove angle — the wedge effect is why V-belts transmit several times the load of a flat belt at equal tension.

F-126

Belt power and centrifugal tension

$$P = \frac{(T_1 - T_2)\, v}{1000} \quad [\text{kW}]; \quad T_c = m'\, v^2 \quad [\text{N}]$$

v belt speed (m/s) · m' belt mass per unit length (kg/m). Centrifugal tension reduces the grip available for power transmission: $T_1 - T_2 = (T_{max} - T_c)(1 - e^{-\mu\theta})$. Above ≈ 30 m/s a standard V-belt loses capacity rapidly — the reason high-speed spindles are direct-driven or use a toothed synchronous belt.

F-127 **Belt length over two pulleys**

$$L = 2C + \frac{\pi}{2}(D + d) + \frac{(D - d)^2}{4C} \quad \text{[mm]}$$

C centre distance · D, d large and small pulley pitch diameters. Solve for C when the belt length is fixed (a standard catalogue length) — this sets the motor-slide position and hence the tensioning adjustment range.

F-128 **Chain speed, tension and polygonal fluctuation**

$$v = \frac{p\,z\,n}{60 \times 10^3} \ \text{[m/s]} \ ; \quad F_t = \frac{1000\,P}{v} \ \text{[N]} \ ; \quad \frac{v_{min}}{v_{max}} = \cos\frac{180°}{z}$$

p chain pitch (mm) · z sprocket teeth · P power (kW). The last term is the polygonal speed ripple — ±3% for a 17-tooth sprocket, ±0.4% for 33 teeth. Chains are therefore unsuitable for feed axes and appear only on conveyors, chip carriers and tool-magazine drives.

WORKED EXAMPLE — A 12-SPEED HEADSTOCK GEARBOX

GIVEN **Speed range to be provided**

n_{min} = 140 rpm, n_{max} = 1800 rpm → R_n = 1800/140 = **12.9** (F-050)

STEP 1 **Choose the number of steps and the progression ratio · F-051, F-053, F-055**

Take z = 12 and the standard R10 ratio φ = 1.26.
Check: $\varphi^{z-1} = 1.26^{11}$ = **12.7** ≈ R_n ✓
Loss of economic cutting speed: A = (1.26 − 1)/1.26 = **20.6%** (F-054)

A finer ratio would reduce the loss but needs more gears: φ = 1.12 (R20) would give A = 10.7% but require z = 1 + ln 12.9/ln 1.12 = 23 speeds. The 20.6% loss is the price of a 12-speed box.

STEP 2 **The twelve speeds · F-052**

Step	1	2	3	4	5	6	7	8	9	10	11	12
calculated	140	176	222	280	353	444	560	705	889	1120	1411	1778
adopted (R20)	**140**	**180**	**224**	**280**	**355**	**450**	**560**	**710**	**900**	**1120**	**1400**	**1800**

Every calculated value rounds onto an R20 preferred number, so the actual φ between adjacent speeds stays within ±3% of 1.26.

STEP 3 **Structural formula and gear count · F-058**

z = 12 = **3(1) · 2(3) · 2(6)**

Three groups → **4 shafts** (motor shaft + three). Group characteristics follow the optimum rule X_i = product of the p values of all preceding groups: 1, 3, 3×2 = 6. No rays cross on the ray diagram, and the last group carries the widest spread, which keeps the intermediate shaft speeds low.
Gear count = 2(p_1 + p_2 + p_3) = 2(3 + 2 + 2) = **14 gears**.

STEP 4 **Ratio-limit check · F-059**

Group 1 (X = 1): i = 1, 1/1.26 = 0.79, $1/1.26^2$ = 0.63 ✓
Group 2 (X = 3): i = 1, $1/1.26^3$ = **0.50** ✓
Group 3 (X = 6): i = 1, $1/1.26^6$ = **0.25** — exactly at the 1/4 sliding-gear limit ✓

This is the fastest-rising structure that is still buildable with sliding clusters. The alternative 2(1)·3(2)·2(6) would need i = $1/1.26^6$ in a group with three sliding gears on one shaft and is rejected on layout grounds.

STEP 5 **Tooth numbers for the first group · F-060, F-061, F-062**

Fix m = 3 mm and Σz = 72 for every pair in the group → centre distance a = 3 × 72/2 = **108 mm**
$z_{driver} = \Sigma z / (1 + 1/i)$:

Pair	required i	z_1 / z_2	actual i	actual φ	error
A (i = 1)	1.000	36 / 36	1.000	–	–
B (i = 1/1.26)	0.794	32 / 40	0.800	1.250	−0.8%
C (i = 1/1.26²)	0.630	28 / 44	0.636	1.572	−1.0%

All pinions have ≥ 28 teeth, comfortably above the 17-tooth undercut limit of F-062, so no profile shift is needed and the centre distance stays at 108 mm for all three pairs. Repeat the procedure for groups 2 and 3 with their own Σz, then verify every one of the twelve output speeds against the adopted R20 values.

GEARBOX PROPORTIONING RULES

Rule	Value or requirement	Formula
Progression ratio φ	1.06 · 1.12 · 1.26 · 1.41 · 1.58 · 2.00 (Renard series)	F-055
Speeds per group p	2 or 3; never more than 4 gears sliding on one shaft	F-058
Group characteristic X_i	product of the p values of all preceding groups	F-058
Group with the widest spread	place **last** — keeps intermediate shaft speeds low and the box compact	F-058
Number of shafts	number of groups + 1 (including the motor shaft)	F-058
Number of gears	$2 \times (p_1 + p_2 + \ldots + p_k)$	F-058
Minimum ratio, sliding cluster	$i \geq 1/4$ — beyond this the two gears cannot share a centre distance	F-059
Maximum ratio, sliding cluster	$i \leq 2$	F-059
Minimum pinion teeth	$z \geq 17$ for $\alpha = 20°$ with no profile shift; ≥ 14 with shift	F-062
Sum of teeth in a group	constant for every pair on the same two shafts: $\Sigma z = 2a/m$	F-061
Centre distance	identical for all pairs in a group; fixed by the largest pair	F-060
Face width	$b = (6 \ldots 10)m$, and $b \leq a/3$	F-119
Speed error after tooth rounding	$\leq \pm 10(\varphi - 1)\%$ at any step, otherwise redistribute Σz	F-052
Gear peripheral speed	sets the required accuracy grade: above 6 m/s use grade 7 or better	F-064
Lubrication	oil bath up to about 12 m/s pitch-line speed; above that, forced circulation	F-064

Why this still matters on a CNC machine. A two-range or three-range gearbox behind a wide-range AC motor is the standard way to obtain full torque at low spindle speed without a very large motor. The range of each stage is designed with exactly the formulas above — typically φ = 2 to 4 with two speeds per group — and the motor's constant-power range (F-105) is then multiplied by the gear ratio. The design check most often missed is the **torque at the lowest spindle speed**: the gearbox must transmit the motor's full stall torque, not merely its rated power.

SECTION 08

Structural stiffness, deflection & joints

F-129 – F-142
14 formulas

Stiffness, not strength, is the governing criterion for almost every machine tool structure. Cutting forces are small compared with what the castings could carry, but the deflection they produce is copied directly into the workpiece.

F-129

Static stiffness

$$K = \frac{F}{\delta} \quad \text{[N/µm]}$$

Always state the **point and direction** of application — a stiffness number without them is meaningless. Machine-tool practice quotes N/µm at the tool centre point (TCP) in the cutting-force direction. Representative targets: small machining centre 15–25 N/µm, medium 25–40 N/µm, heavy-duty and turning 40–100 N/µm, grinding machines 60–150 N/µm.

F-130

Combination of stiffnesses

$$\text{series: } \frac{1}{K} = \Sigma \frac{1}{K_i} \quad ; \quad \text{parallel: } K = \Sigma K_i$$

The load path of a machine is a series chain: tool → holder → spindle → bearings → headstock → column → bed → guideway → table → workpiece. The **softest** element dominates; adding stiffness anywhere else buys almost nothing. Build the chain, then spend money on the weakest link.

F-131

Machine loop (tool–workpiece) stiffness

$$K_{\text{loop}} = \frac{F}{\delta_{\text{loop}}} \quad , \quad \delta_{\text{loop}} = \delta_{\text{tool}} + \delta_{\text{spindle}} + \delta_{\text{struct}} + \delta_{\text{guide}} + \delta_{\text{fix}} + \delta_{\text{work}}$$

The relative displacement between cutting edge and workpiece under cutting force — the quantity that actually determines dimensional error and chatter resistance. The workpiece and fixture are frequently the softest members and are the ones the machine designer controls least.

F-132

Dimensional error produced by compliance

$$\Delta = \frac{F}{K_{\text{loop}}} \quad \text{[µm]}$$

In turning this appears as a diameter error of 2Δ (the tool is pushed away by F_p); in milling as a size and form error that follows the force direction. Because F varies along the cut, the error varies too — producing taper, barrel shape and out-of-roundness rather than a simple offset that could be compensated.

F-133

Beam deflections — standard cases

$$\text{cantilever, tip load: } \delta = \frac{F L^3}{3E I} \quad ; \quad \text{cantilever, u.d.l.: } \delta = \frac{w L^4}{8E I}$$

$$\text{simply supported, centre load: } \delta = \frac{F L^3}{48E I} \quad ; \quad \text{simply supported, u.d.l.: } \delta = \frac{5w L^4}{384E I}$$

w load per unit length (N/mm). The cube law on span is the designer's most powerful lever: a column 20% taller is 73% more flexible unless the section is also deepened. Machine beds are simply-supported or multi-point-supported beams; columns and cross-rails are cantilevers.

F-134

End rotation of a cantilever

$$\theta = \frac{F L^2}{2E I} \quad \text{[rad]}$$

Multiply by the Abbe offset to convert a structural rotation into a linear error at the TCP (F-224). A 20 µrad tilt of a spindle head with a 400 mm Abbe offset produces 8 µm of position error.

F-135

Torsional deformation

$$\theta = \frac{T L}{G J} \quad \text{[rad]} \; ; \quad J_{\text{solid}} = \frac{\pi d^4}{32} \quad ; \quad J_{\text{hollow}} = \frac{\pi (D^4 - d^4)}{32}$$

G = 80 GPa for steel, 26–45 GPa for cast iron depending on grade. Torsion of the bed and of the column is usually the dominant compliance of a machining centre under the overturning moment of an off-centre cut.

F-136

Torsional stiffness

$$K_t = \frac{G J}{L} \quad \text{[N·m/rad]}$$

Used both for static twist and as the spring in the torsional natural frequency of a drive train (F-180). A closed (box) section has a far larger J than an open one of equal mass — always close the box.

F-137

Second moment of area — common machine sections

$$\text{rectangle: } I = \frac{b\,h^3}{12} \;;\quad \text{hollow rectangle: } I = \frac{b\,h^3 - b_1 h_1{}^3}{12} \;;\quad \text{circle: } I = \frac{\pi\,d^4}{64}$$

Section modulus $Z = I/y_{max}$. Depth h is cubed, width b is linear: material belongs as far from the neutral axis as the design allows, which is why machine structures are deep, thin-walled and ribbed.

F-138

Joint (contact) stiffness

$$K_j = c\,A\,p^m \quad \text{[N/µm]}$$

A apparent contact area (cm²) · p nominal contact pressure (N/mm²) · $m \approx 0.5$–1.0 · c a constant set by surface finish and joint type. Order of magnitude for machined cast-iron joints at 1–3 MPa: 5–50 N/µm per cm²; scraped and damped joints lower stiffness but far higher damping. Bolted joints commonly account for **50–80% of the total compliance** of a machine structure — always model them, never assume them rigid. Determine c and m by test on the actual joint.

F-139

Damping capacity and damping ratio

$$\psi = \frac{\Delta U}{U} \;;\quad \zeta = \frac{\psi}{4\pi}$$

ψ specific damping capacity = energy dissipated per cycle ÷ maximum strain energy. Grey cast iron ψ ≈ 3–10× that of welded steel at equal stress, which is why beds and columns are cast; welded steel structures recover the difference with sand, concrete or polymer filling and with damped joints.

F-140

Stiffness required by the accuracy specification

$$K_{req} \geq \frac{F_{max}}{\delta_{allow}}$$

F_{max} maximum cutting force at full power and lowest speed · δ_{allow} the share of the total error budget allocated to static deflection (typically one third of the geometric tolerance). This inequality, evaluated before any geometry is drawn, is the single most useful line in the whole handbook.

F-141

Stiffness scaling with wall thickness

$$\text{bending: } K \propto t^3 \;;\quad \text{shear: } K \propto t \;;\quad \text{mass: } m \propto t$$

Doubling a wall thickness multiplies bending stiffness by 8 for only twice the mass — until shear deformation and joint compliance take over, typically beyond $t \approx 25$–40 mm for cast iron. Past that point add ribs and depth instead of thickness.

F-142

Machine mass and foundation pressure

$$m = \rho\,V \;;\quad p_f = \frac{m\,g}{A_f} \leq p_{allow}$$

ρ: grey cast iron 7200, steel 7850, mineral casting / polymer concrete 2200–2400, aluminium 2700 kg/m³. A_f total levelling-pad or foundation area. Typical floor limits: 0.15–0.30 N/mm² point load under a pad, 0.02–0.05 N/mm² distributed. A useful design ratio: machine mass ÷ maximum workpiece mass of 3–5 for general machining, 8–15 for precision grinding.

SECTION 09

Strength, fatigue & machine elements

F-143 – F-170
28 formulas

The general mechanical design set, applied the way it is applied on a machine tool: shafts, keys, bolts, welds, press fits, springs and contact stresses — checked against fatigue, not against yield.

STRESS AND STRAIN

F-143

Direct stress, strain and Hooke's law

$$\sigma = \frac{F}{A} \;;\quad \varepsilon = \frac{\Delta L}{L} \;;\quad \sigma = E\,\varepsilon$$

E: 207 GPa steel, 100–140 GPa grey cast iron, 70 GPa aluminium, 15–40 GPa mineral casting. A useful design fact: at 100 N/mm² a steel member strains 483 µm/m — so a 1 m bed carrying modest stress already moves more than most accuracy budgets allow.

F-144

Elastic constant relation

$$G = \frac{E}{2(1+\nu)}$$

ν Poisson's ratio: 0.29–0.30 steel, 0.21–0.27 grey cast iron, 0.33 aluminium. Gives $G \approx 80$ GPa for steel and ≈ 44 GPa for cast iron — cast iron is proportionally much weaker in torsion than in tension.

F-145

Axial elongation

$$\Delta L = \frac{F\,L}{A\,E} \quad \text{[mm]}$$

For a stepped member sum the segments: $\Delta L = \Sigma\, F_i\, L_i/(A_i\, E)$. This is the same expression as the screw-shaft stiffness of F-076.

F-146

Bending stress and section moduli

$$\sigma = \frac{M\,y}{I} = \frac{M}{Z} \;;\quad Z_{\text{rect}} = \frac{b\,h^2}{6} \;;\quad Z_{\text{circ}} = \frac{\pi\,d^3}{32}$$

M bending moment (N·mm) at the section considered. For a cantilever $M = F\,L$; for a simply supported beam with a centre load $M = F\,L/4$.

F-147

Torsional shear stress

$$\tau = \frac{T\,r}{J} = \frac{T}{Z_{\text{p}}} \;;\quad Z_{\text{p,circ}} = \frac{\pi\,d^3}{16}$$

Twice the section modulus of the bending case for the same diameter — a shaft is proportionally stronger in torsion than in bending.

F-148

Von Mises equivalent stress

$$\sigma_{\text{v}} = \sqrt{\sigma^2 + 3\tau^2} = \sqrt{\frac{1}{2}\left[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2\right]}$$

The correct yield criterion for ductile steel and cast aluminium. For a shaft in combined bending and torsion this reduces to $\sigma_{\text{v}} = \sqrt{(\sigma_{\text{b}}^2 + 3\tau_{\text{t}}^2)}$. Grey cast iron is brittle — use the maximum-principal-stress criterion (F-149) for it instead.

F-149

Principal stresses and maximum shear stress

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^{\,2}} \;;\quad \tau_{\max} = \frac{\sigma_1 - \sigma_3}{2}$$

Use the principal-stress criterion for brittle cast-iron beds and columns, where the tensile principal stress — not the von Mises value — governs cracking.

F-150

Factor of safety

$$n = \frac{\sigma_{\text{lim}}}{\sigma_{\text{work}}}$$

σ_{lim} = yield (static ductile), ultimate (static brittle) or modified endurance limit (fatigue). Machine-tool practice: 1.5–2.0 on yield for well-understood static loads; 2.5–4 for anything subject to crash loads, clamping forces or impact; 3–4 on buckling (F-072). Remember that a machine is far more often limited by **deflection** than by stress — check F-140 first.

F-151

Stress concentration and notch sensitivity

$$\sigma_{max} = K_t \, \sigma_{nom} \; ; \quad K_f = 1 + q(K_t - 1)$$

K_t theoretical (geometric) concentration factor · K_f fatigue notch factor · q notch sensitivity 0–1. Typical K_t: sharp shoulder fillet 2.5–4, keyway 1.6–2.0 (profile milled) to 2.5+ (sunk), transverse hole 2.0–2.5, thread root 2.5–3.5. Use K_f in fatigue, K_t only for brittle materials.

BUCKLING AND FATIGUE

F-152

Slenderness ratio and column failure mode

$$\lambda = \frac{L_e}{\rho_g} \; , \quad \rho_g = \sqrt{I/A} \; ; \quad \lambda_c = \sqrt{\frac{2\pi^2 E}{\sigma_y}}$$

$\lambda > \lambda_c$ Euler: $\sigma_{cr} = \dfrac{\pi^2 E}{\lambda^2}$; $\lambda \le \lambda_c$ Johnson: $\sigma_{cr} = \sigma_y - \dfrac{1}{E}\left(\dfrac{\sigma_y}{2\pi}\right)^2 \lambda^2$

$L_e = K\,L$ effective length with the end-fixity factors of F-071 · ρ_g radius of gyration. For structural steel $\lambda_c \approx 125$; for cast iron ≈ 80. Short machine columns fail by crushing, slender screw shafts and rams by buckling.

F-153

Fatigue criteria for combined mean and alternating stress

Goodman $\dfrac{\sigma_a}{\sigma_e} + \dfrac{\sigma_m}{\sigma_{UTS}} = \dfrac{1}{n}$; Soderberg $\dfrac{\sigma_a}{\sigma_e} + \dfrac{\sigma_m}{\sigma_y} = \dfrac{1}{n}$; Gerber $\dfrac{n\,\sigma_a}{\sigma_e} + \left(\dfrac{n\,\sigma_m}{\sigma_{UTS}}\right)^2 = 1$

σ_a alternating, σ_m mean stress, σ_e modified endurance limit. Goodman is the general machine-design default; Soderberg is more conservative and suits ductile steel with high mean stress (preloaded bolts, pre-tensioned screws); Gerber is least conservative and needs good material data.

F-154

Modified endurance limit

$$\sigma_e = k_a \, k_b \, k_c \, k_d \, k_e \, \sigma'_e \; , \quad \sigma'_e \approx 0.5 \, \sigma_{UTS}$$

k_a surface finish (0.7–1.0; ground 0.95, machined 0.85, as-cast 0.6) · k_b size (≈ 1.0 below 8 mm, falling to ≈ 0.75 at 50 mm) · k_c load type (1.0 bending, 0.70 axial, 0.59 torsion) · k_d temperature · k_e reliability (0.814 for 99.9%). The $0.5\sigma_{UTS}$ rule holds for steels up to about 1400 N/mm²; grey cast iron has no true endurance limit and is designed on a damped σ'_e of $0.35–0.45\sigma_{UTS}$.

F-155

Cumulative fatigue damage (Miner's rule)

$$D = \Sigma \, \frac{n_i}{N_i} \le 1$$

n_i cycles applied at stress level i · N_i cycles to failure at that level from the S–N curve. Use it to combine the different load cases a machine sees: rapid traverse, cutting, tool change, clamping and crash. Load sequence is ignored — a known limitation, acceptable for design but not for life extension of a cracked part.

FASTENERS, KEYS AND SHAFTS

F-156

Bolt preload

$$F_0 = (0.60 \ldots 0.75) \, \sigma_y \, A_s \quad [\text{N}]$$

High preload is **desirable** in machine-tool joints: it keeps the joint closed under load, raises joint stiffness (F-138) and improves damping. Use the upper end for permanent, torque-controlled joints and the lower end where the bolt is dismantled in service.

F-157

Bolt tightening torque

$$T = K \, F_0 \, d \quad [\text{N·m}]$$

d nominal bolt diameter (m) · K nut factor ≈ 0.20 dry, 0.15 lightly oiled, 0.10–0.12 with MoS_2 paste. K varies ±30% with surface condition, so torque control alone gives poor preload accuracy — for critical joints specify preload directly and use a hydraulic tensioner, ultrasonic measurement or turn-of-nut.

F-158

Bolt tensile stress area

$$A_s = \frac{\pi}{4} \left(\frac{d_2 + d_3}{2}\right)^2$$

d_2 pitch diameter, d_3 minor diameter of the thread. For M8 ≈ 36.6 mm², M10 ≈ 58.0 mm², M12 ≈ 84.3 mm², M16 ≈ 157 mm², M20 ≈ 245 mm² (ISO 898 metric coarse).

F-159

Joint load factor and bolt load under external force

$$\Phi = \frac{K_b}{K_b + K_j} \;;\quad F_{bolt} = F_0 + \Phi\, F_{ext} \;;\quad F_{joint} = F_0 - (1 - \Phi) F_{ext}$$

K_b, K_j bolt and clamped-member stiffness. Φ is typically 0.1–0.3 for a stiff cast-iron joint: only a small fraction of the external load reaches the bolt, provided the joint stays closed. If F_{joint} reaches zero the joint separates, the bolt takes 100% of the load and fatigue life collapses — check separation before checking bolt stress.

F-160

Key strength — shear and crushing

$$\tau = \frac{2T}{d\, w\, L} \le \tau_{all} \;;\quad \sigma_c = \frac{4T}{d\, h\, L} \le \sigma_{c,all}$$

d shaft diameter · w key width · h key height · L engaged length. Crushing usually governs; a square key of side ≈ $d/4$ with length ≈ $1.5d$ is approximately equally strong in shear and crushing and is the standard starting proportion.

F-161

Shaft diameter for combined bending and torsion (ASME code form)

$$d = \left[\frac{16}{\pi\, \tau_{all}} \sqrt{(K_m\, M)^2 + (K_t\, T)^2} \right]^{1/3} \quad \text{[mm]}$$

K_m, K_t combined shock-and-fatigue factors: 1.5 for a gradually applied load, 2.0 for sudden light shock, 2.5–3.0 for heavy shock. $\tau_{all} \approx 0.3\sigma_y$ for a shaft without keyways, 0.18–0.24σ_y with them. Take M and T at the same critical section, then increase d for the keyway and check deflection — a machine-tool shaft is nearly always sized by stiffness, not by this equation.

F-162

Combined critical speed of a multi-mass shaft (Dunkerley)

$$\frac{1}{n_c^2} = \frac{1}{n_1^2} + \frac{1}{n_2^2} + \ldots + \frac{1}{n_k^2}$$

n_i critical speed with mass i acting alone. Dunkerley gives a **lower** bound (conservative) and is accurate to a few percent when one mass dominates. Use it for spindles carrying a chuck and a workpiece, and for feed screws carrying the nut and the table.

F-163

Fillet-weld throat stress

$$\tau = \frac{F}{0.707\, h\, L} \le \tau_{all} \;;\quad \sigma = \frac{M}{0.707\, h\, L^2/6} \text{ (bending, per unit run)}$$

h weld leg size · L effective length · 0.707 = 1/√2 converts leg to throat. $\tau_{all} \approx 0.3 \times$ electrode tensile strength for static load, and roughly half that for fatigue. Welded machine frames are fatigue-critical at weld ends and attachments — specify a ground weld toe and avoid transverse-loaded fillets.

FITS AND CONTACT

F-164

Press-fit contact pressure (Lamé)

$$p = \frac{\delta}{d\left[\frac{1}{E_o}\left(\frac{d_o^2 + d^2}{d_o^2 - d^2} + \nu_o\right) + \frac{1}{E_i}\left(\frac{d^2 + d_i^2}{d^2 - d_i^2} - \nu_i\right)\right]}$$

δ total diametral interference · d interface diameter · d_o hub outside diameter · d_i shaft bore (0 if solid) · subscripts o = outer member (hub), i = inner member (shaft). Used for bearing seats, shrink-fit tool holders, gear hubs and pre-tensioned screw nuts. Check the hub hoop stress $\sigma_\theta = p(d_o^2 + d^2)/(d_o^2 - d^2)$ against the hub material.

F-165

Press-fit torque and axial capacity

$$T_{cap} = \frac{\pi}{2}\, \mu\, p\, d^2\, L \;;\quad F_{cap} = \pi\, \mu\, p\, d\, L$$

L engagement length · $\mu \approx 0.12$ pressed steel/steel, 0.08–0.10 for a shrink fit after heating (lubricated by the oxide film). This is the same form as F-115 for a taper connection.

F-166

Thermal stress in a constrained member

$$\sigma_{th} = E\, \alpha\, \Delta T \quad \text{[N/mm²]}$$

Fully restrained steel develops 2.4 N/mm² per kelvin — a 40 K gradient gives 97 N/mm², enough to yield mild steel and to crack a cast-iron bed. This is why machine beds are ribbed to be flexible in the through-thickness direction and why weldments are stress-relieved.

F-167

Hertzian contact — sphere on surface (point contact)

$$a = \left(\frac{3F\,R}{4E^*}\right)^{1/3} \;;\quad p_{\max} = \frac{3F}{2\pi\,a^2} \;;\quad \frac{1}{E^*} = \frac{1-\nu_1^{\,2}}{E_1} + \frac{1-\nu_2^{\,2}}{E_2}$$

R equivalent radius, $1/R = 1/R_1 + 1/R_2$ (negative for a concave raceway). Governs ball-to-raceway stress in bearings and ball screws (typically 2000–4200 N/mm², below the ≈ 4200 N/mm² static limit for through-hardened bearing steel) and ball-to-groove stress in linear guides.

F-168

Hertzian contact — cylinders (line contact)

$$b = \sqrt{\frac{4F\,R}{\pi\,L\,E^*}} \;;\quad p_{\max} = \frac{2F}{\pi\,b\,L} = \sqrt{\frac{F\,E^*}{\pi\,L\,R}}$$

L contact length. Applies to roller bearings, gear teeth at the pitch point, wheel/rail and cam followers. Contact stress falls only with the square root of load — doubling the load raises p_{max} by 41%.

F-169

Helical compression spring

$$k = \frac{G\,d^4}{8D^3N} \;;\quad \tau = \frac{8F\,D\,K_w}{\pi\,d^3} \;;\quad K_w = \frac{4C-1}{4C-4} + \frac{0.615}{C}$$

d wire diameter · D mean coil diameter · N active coils · $C = D/d$ spring index, keep 4–12 · K_w Wahl curvature-correction factor. Used for tool-clamping disc springs, counterbalances, accumulator springs and way wipers. Disc (Belleville) springs — the usual choice for drawbars — stack in series/parallel and follow their own catalogue curves.

F-170

Bearing pressure and *pv* limit on sliding surfaces

$$p = \frac{F}{b\,d} \le p_{\text{all}} \;;\quad p\,v \le (p\,v)_{\text{all}}$$

b bearing length · d journal diameter. The pv product limits frictional heat generation and is the governing criterion for plain bearings, way surfaces, bushings and clamping elements. Typical limits: bronze 1.8–3.5 N/mm²·m/s, sintered bronze 0.5–1.8, PTFE-lined 0.3–1.0, hydrodynamically lubricated white metal up to 10.

SECTION 10

Dynamics, vibration & chatter stability

F-171 – F-184
14 formulas

A structure that is stiff enough statically can still be unusable dynamically. These formulas give the natural frequencies, the response to forced excitation, the isolation design and the regenerative-chatter stability limit that sets the maximum usable depth of cut.

FREE AND FORCED VIBRATION

F-171

Natural frequency of a single-degree-of-freedom system

$$f_n = \frac{1}{2\pi}\sqrt{\frac{K}{m}} = \frac{1}{2\pi}\sqrt{\frac{g}{\delta_{st}}} \quad \text{[Hz]}$$

K stiffness (N/m) · m mass (kg) · δ_{st} static deflection under the weight (m). The second form is the fastest estimate available: measure or compute the static sag and you have the first natural frequency. Machine structures target a first mode above 80–150 Hz so that it lies clear of tooth-passing frequencies at normal cutting speeds.

F-172

Damping ratio from the logarithmic decrement

$$\delta_{log} = \frac{1}{k}\ln\frac{x_1}{x_{1+k}} \;;\; \zeta = \frac{\delta_{log}}{\sqrt{4\pi^2 + \delta_{log}^2}} \approx \frac{\delta_{log}}{2\pi}$$

Obtained from a hammer-impact (bump) test on the assembled machine. Typical machine-tool values: ζ = 0.01–0.03 for a bolted cast-iron structure, 0.03–0.08 with damped joints or polymer filling, 0.005–0.01 for a welded steel frame with no damping treatment.

F-173

Quality factor and resonant amplitude

$$Q_d = \frac{1}{2\zeta} \;;\; X_{res} = \frac{F_0}{2\zeta K} = \frac{F_0}{c\,\omega_n}$$

F_0 harmonic force amplitude · $c = 2\zeta\sqrt{(Km)}$ viscous damping coefficient. At ζ = 0.02 the resonant amplitude is 25× the static deflection — the reason a machine that runs smoothly at 3000 rpm can destroy a tool at 3150.

F-174

Dynamic magnification factor

$$M = \frac{X}{X_{st}} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \quad r = \frac{\omega}{\omega_n}$$

$X_{st} = F_0/K$. $M \to 1$ for $r \ll 1$ (quasi-static), peaks at $r = \sqrt{(1-2\zeta^2)}$, and $\to 1/r^2$ for $r \gg 1$ (mass-controlled). Plot M against the tooth-passing frequency sweep of the intended cutting range before releasing the design.

ISOLATION

F-175

Vibration transmissibility

$$TR = \sqrt{\frac{1 + (2\zeta r)^2}{(1-r^2)^2 + (2\zeta r)^2}}$$

Ratio of transmitted to applied force (or of response to base excitation). TR > 1 for $r < \sqrt{2}$ — below that frequency ratio the mount **amplifies** rather than isolates, and every start-up and shut-down passes through that band.

F-176

Isolation efficiency

$$\eta_{iso} = (1 - TR) \times 100 \quad [\%], \quad \text{requires } r > \sqrt{2}$$

Design mounts for r = 2.5–5, giving 80–96% isolation. Note that isolation protects the machine from floor vibration and the floor from the machine — a precision grinder needs both, and its mount must be stiff enough in the vertical axis not to compromise geometric accuracy under load.

F-177

Mount natural frequency from static deflection

$$f_n \approx \frac{15.76}{\sqrt{\delta_{st}}} \quad [\text{Hz, } \delta_{st} \text{ in mm}] \;;\; \delta_{st} = \frac{m\,g}{k}$$

A 1 mm static deflection gives 15.8 Hz; 4 mm gives 7.9 Hz. Distribute the machine weight over the pads so each carries its design share — an unlevelled machine loads one pad and unloads the others, changing f_n and the geometry at the same time.

TORSIONAL AND SHAFT DYNAMICS

F-178

Torsional natural frequency

$$f_t = \frac{1}{2\pi}\sqrt{\frac{K_t}{J}}\ , \quad K_t = \frac{G J_p}{L} \quad \text{[Hz]}$$

The torsional mode of a ball screw between motor and nut is usually 100–300 Hz and is the mode that limits the achievable position-loop gain (F-095). Compute it before setting servo gains, not after a vibration complaint.

F-179

Two-mass torsional system

$$f_t = \frac{1}{2\pi}\sqrt{K_t\left(\frac{J_1 + J_2}{J_1 J_2}\right)}$$

Motor inertia J_1, load inertia J_2, shaft torsional stiffness K_t. The bracketed term is the reciprocal of the reduced (equivalent) inertia. Keep this frequency at least 3–4× the position-loop bandwidth, or the loop will excite it.

F-180

Shaft critical speed from static deflection (Rayleigh)

$$n_c = 60\, f_n \approx \frac{60 \times 0.5625}{\sqrt{\delta_{st}}} \quad \text{[rpm, } \delta_{st} \text{ in m]}$$

δ_{st} static deflection of the shaft under its own weight (simply supported: $\delta = 5wL^4/384EI$). Equivalent to the beam-theory result $n_c = (60/2\pi)(\lambda L)^2\sqrt{(EI/\rho A)}/L^2$ and the basis of the catalogue constants in F-069. Operate below $0.75n_c$ or above $1.25n_c$, never in between.

CHATTER

F-181

Regenerative chatter — limiting depth of cut

$$a_{p,lim} = -\frac{1}{2\, k_f\, \mathrm{Re}[G(\omega_c)]} \quad \text{[mm]}$$

k_f specific cutting force per unit width of cut (N/mm²) · $G(\omega_c)$ direct frequency-response function at the tool tip (mm/N), measured by a hammer test or computed by FEA · Re[] its real part, which is negative above each natural frequency — hence the minus sign. **The single most useful dynamic result in machine design:** the stable depth of cut is inversely proportional to the negative real part of the FRF, so raising stiffness and damping at the dominant mode raises the productive cutting window directly.

F-182

Chatter lobe — spindle speeds at a given chatter frequency

$$\Omega = \frac{60\ \omega_c}{z(\varepsilon + 2k\pi)} \quad \text{[rpm]}$$

ω_c chatter frequency (rad/s) · z number of teeth · ε phase shift between the inner and outer surface waviness (rad) · k = 0, 1, 2 … lobe number. Plotting F-181 against F-182 for a sweep of ω_c produces the stability-lobe diagram: alternating pockets of stable depth of cut at particular spindle speeds. Choosing a speed in a lobe pocket can multiply the stable depth of cut several times with no hardware change.

F-183

Tooth-passing frequency and force pulsation

$$f_t = \frac{z\, n}{60} \quad \text{[Hz]}\ ; \quad \Delta F \approx F_{c,mean}\left(1 - \frac{1}{z_e}\right)$$

The excitation frequency the structure must not coincide with. With $z_e < 1$ the force pulses between zero and peak every tooth pass — check f_t and its harmonics against the modes from F-171 across the whole speed range, and vary tooth count or speed to move the excitation off a mode.

F-184

Impact and shock load factor

$$F_{imp} = F_{st}\left(1 + \sqrt{1 + \frac{2h}{\delta_{st}}}\right)\ ; \quad E_{imp} = \frac{1}{2} m\, v^2$$

h drop height · δ_{st} static deflection under F_{st}. A stiff structure sees a **larger** impact force for the same energy — the argument for controlled compliance and for energy-absorbing elements in crash protection. Use it to size the crash load case for guideways (F-082), screws (F-072) and spindle bearings.

SECTION 11

Guideways — sliding, rolling, hydrostatic & aerostatic

F-185 – F-194
10 formulas

The guideway carries the cutting force, defines the geometric accuracy of the axis and, in sliding form, determines whether the motion is smooth at 1 mm/min or stick-slips.

SLIDING AND ROLLING GUIDEWAYS

F-185

Friction and total drive force

$$F_f = \mu\, W\ ; \quad F_{drive} = F_{cut} + \mu\, W + m\, a + F_{seal} + F_{pre}$$

μ: 0.002–0.005 recirculating ball/roller guides, 0.005–0.02 hydrostatic, 0.005–0.01 aerostatic, 0.05–0.10 PTFE-lined sliding ways, 0.10–0.20 cast iron on hardened steel with way lubrication, 0.15–0.30 dry or starved. F_{pre} is the drag from guide preload and wipers, often 200–800 N on a loaded axis and easily the largest term at low speed.

F-186

Way bearing pressure

$$p = \frac{W}{A} \le p_{all}$$

Typical allowable pressures for continuously lubricated sliding ways: cast iron on hardened steel 1.5–3.0 N/mm², PTFE-lined 2.0–4.0 N/mm², hydrostatic 3.0–7.0 N/mm². Exceeding it squeezes out the lubricant film, accelerates wear and destroys geometric accuracy permanently — wear, not fracture, is the failure mode of a slideway.

F-187

Maximum edge pressure with an overturning moment

$$p_{max} = \frac{W}{A} + \frac{M}{Z} = \frac{W}{A}\left(1 + \frac{6e}{B}\right) \quad [\text{N/mm}^2]$$

M overturning moment · Z section modulus of the way contact area · $e = M/W$ eccentricity of the load · B way width. When $e > B/6$ the load leaves the middle of the way and pressure concentrates at the edge — the reason long overhung saddles wear at their ends first. Keep e/B below 1/6 wherever the layout allows.

F-188

Stick-slip avoidance criterion

$$K_{drive} \ge \frac{(\mu_s - \mu_d)\, W}{x_s} \quad [\text{N/mm}]$$

μ_s, μ_d static and dynamic friction · x_s elastic displacement needed to break the joint free (typically 0.01–0.05 mm). Stick-slip appears when the drive is **softer** than the friction drop: the axis creeps at low speed and jumps at the break-away point, ruining fine feeds and surface finish. Cure it by raising drive stiffness (pre-tensioned screw, direct drive), reducing the friction drop (PTFE-lined ways, hydrostatic, way lubricant with friction modifiers) or both.

HYDROSTATIC AND AEROSTATIC GUIDEWAYS AND BEARINGS

F-189

Hydrostatic pad load capacity

$$W = p_r\, A_e\ ; \quad A_e = \text{effective pad area} \quad [\text{mm}^2]$$

p_r recess pressure (N/mm²). A_e is the pad area reduced by the recess and weighted for the land pressure distribution — for a rectangular pad with a central recess, $A_e \approx A_{pad}\cdot(1 + \text{recess fraction})/2$ to a good approximation, and is tabulated exactly in the design handbook for the pad geometry used. Set $p_r/p_s = \beta \approx 0.5$ for capillary compensation, 0.6–0.7 for orifice compensation.

F-190

Oil flow through a bearing land (parallel plates)

$$Q = \frac{p_r\, b\, h^3}{12\, \mu\, l} \quad [\text{mm}^3/\text{s}]$$

b land width across the flow · l land length along the flow · h film thickness · μ dynamic viscosity (N·s/mm²). Sum over all lands of the pad. The cubic dependence on h is the whole design principle: halving the film octuples the flow and the pump power, and roughly doubles the stiffness.

F-191

Hydrostatic pad stiffness

$$K = \frac{3\, p_r\, A_e}{h}\left(1 - \frac{1}{3\beta}\right) \quad [\text{N/mm}]$$

Constant-supply (ideal) case gives $K = 3p_rA_e/h$; the bracket corrects for a fixed restrictor with pressure ratio $\beta = p_r/p_s$, maximised near $\beta \approx 0.5$–0.6. A hydrostatic way reaches 200–1000 N/µm — an order of magnitude above a rolling guide — with zero static friction and infinite life, at the cost of a hydraulic power unit, oil cooling and considerably more design work.

F-192

Hydrostatic pump power and oil temperature rise

$$P_{\text{pump}} = \frac{p_{\text{s}}\, Q}{600\, \eta} \quad \text{[kW; } p_{\text{s}} \text{ in bar, } Q \text{ in L/min]} \quad ; \quad \Delta T_{\text{oil}} = \frac{P_{\text{loss}}}{\dot{m}\, c_{\text{p}}}$$

Nearly all pump power becomes heat in the oil. Mineral hydraulic oil: $\rho \approx 870$ kg/m³, $c_{\text{p}} \approx 1.9$ kJ/(kg·K). Hold the supply temperature to ±0.5 K on a precision machine — the oil film thickness and therefore the axis position follow it directly.

F-193

Aerostatic pad load, stiffness and air consumption

$$W = p_{\text{r}}\, A_{\text{e}} \quad ; \quad K \approx \frac{p_{\text{r}}\, A_{\text{e}}}{h} \quad ; \quad Q_{\text{air}} = \frac{p_{\text{r}}\, b\, h^{3}}{12\, \mu\, l}$$

Same relations with air ($\mu \approx 1.8 \times 10^{-11}$ N·s/mm² at 20 C, ~50× less viscous than oil) and a compressibility correction. Load capacity is limited by the practical supply pressure of 0.4–0.7 MPa, so aerostatic ways suit light, fast, ultra-precision machines (diamond turning, measuring machines, PCB routers) where friction-free motion matters more than load capacity.

F-194

Rolling guideway friction power and heat

$$P_{\text{loss}} = \mu\, W\, v \quad \text{[W]} \quad ; \quad \mu = \mu_{0} + \frac{c}{v}$$

v carriage speed (m/s). Rolling friction is low but not zero, and it rises with speed through ball-cage and recirculation losses. On a high-speed machine the guide friction heat is comparable to the screw heat and must enter the thermal model (F-195) — it is what makes the column and bed bow during a long high-speed run.

SECTION 12

Thermal behaviour & coolant systems

F-195 – F-204
10 formulas

On a modern machine tool, thermal effects cause more lost accuracy than every other source combined. These formulas quantify the heat generated, the temperature reached, the growth produced and the coolant flow needed to control it.

F-195

Heat generated by cutting and its partition

$$\dot{Q} = P_c \quad [\mathrm{W}] \; ; \quad \dot{Q}_{\mathrm{chip}} \approx (0.60 \ldots 0.80)\,\dot{Q} \; , \quad \dot{Q}_{\mathrm{work}} \approx (0.10 \ldots 0.25)\,\dot{Q} \; , \quad \dot{Q}_{\mathrm{tool}} \approx (0.03 \ldots 0.10)\,\dot{Q}$$

Essentially all cutting power becomes heat. The fractions shift with speed (more to the chip), with feed and with coolant effectiveness. The share entering the **workpiece and the machine** — 15–30% — is what drives thermal error; at 20 kW of cutting power that is 3–6 kW poured into the structure.

F-196

Conduction and thermal resistance

$$\dot{Q} = k\,A\,\frac{\Delta T}{L} \; ; \quad R_{\mathrm{th}} = \frac{L}{k\,A} \quad [\mathrm{K/W}]$$

k thermal conductivity: 50 W/(m·K) steel, 48–55 grey cast iron, 220 aluminium, 1.0–1.5 mineral casting, 0.15 oil film. Mineral casting's very low k is an advantage for thermal **stability** (slow, damped response) and a disadvantage for removing heat from a mounting surface.

F-197

Convection

$$\dot{Q} = h\,A\,\Delta T \quad [\mathrm{W}]$$

h: 5–10 W/(m²·K) free convection over a machine surface, 10–50 forced air, 200–1000 flood coolant, 5 000–20 000 oil-jet or through-spindle cooling at the cutting zone, 10 000–50 000 for a jet aimed directly at the cut. Getting the coolant to the cutting edge — not the flow rate — is what determines which of these applies.

F-198

Steady-state temperature rise of a component

$$\Delta T = \dot{Q}\,R_{\mathrm{th}} = \frac{\dot{Q}}{h\,A} \quad [\mathrm{K}]$$

Build the machine as a resistance network: bearing heat → housing → column → bed → floor, and cutting heat → coolant → tank → chiller. Every interface adds a contact resistance that is usually larger than the conduction resistance of the metal itself.

F-199

Coolant flow required for heat removal

$$\dot{m} = \frac{\dot{Q}}{c_p\,\Delta T} \quad [\mathrm{kg/s}] \; ; \quad \dot{V} = \frac{\dot{m}}{\rho} \quad [\mathrm{L/min} = 60\,\dot{V} \times 10^3]$$

Water-based coolant: $\rho \approx 1000$ kg/m³, $c_p \approx 3.9$ kJ/(kg·K) for a 5% emulsion. Removing 5 kW with a 3 K rise needs ≈ 25 L/min at the cut — but the tank and chiller must handle the total, so specify both the delivery rate and the return capacity.

F-200

Coolant temperature rise through the cut

$$\Delta T_{\mathrm{coolant}} = \frac{\dot{Q}_{\mathrm{to\ coolant}}}{\dot{m}\,c_p} \quad [\mathrm{K}]$$

If the coolant is recirculated without chilling, the whole machine soaks to a rising temperature and the structure grows continuously. On a machine specified for ±5 µm over 8 h, coolant temperature control to ±0.5 K is normally mandatory.

F-201

Thermal growth of a machine element

$$\Delta = \alpha\,L\,\Delta T \quad [\mathrm{mm}] \; ; \quad \Delta Z_{\mathrm{TCP}} = \Sigma\,\alpha_i\,L_i\,\Delta T_i$$

Sum along the whole structural loop from the scale reference to the cutting edge. Cast iron $\alpha = 10.5 \times 10^{-6}$/K, steel 11.7×10^{-6}/K, aluminium 23×10^{-6}/K, granite 6×10^{-6}/K, mineral casting $10–18 \times 10^{-6}$/K. A 1 m column at +3 K moves the TCP 35 µm.

F-202

Thermal time constant and warm-up curve

$$\tau_{\mathrm{th}} = \frac{m\,c_p}{h\,A} \; ; \quad T(t) = T_\infty\,(1 - \mathrm{e}^{-t/\tau_{\mathrm{th}}})$$

Steady state is reached after ≈ $3–5\tau_{\mathrm{th}}$. Machine tool thermal time constants run 2–12 h, so a machine is never thermally settled during a normal shift — which is why warm-up cycles and thermal compensation, not thermal equilibrium, are the practical design answers.

F-203

Thermal error compensation

$$\Delta Z_{comp} = \Sigma\, \alpha_i\, L_i\, (T_i - T_{ref}) \quad \text{[mm]}$$

T_{ref} = 20 C by convention (ISO 1). Requires temperature sensors at each structural element, a calibrated model of the $\alpha_i L_i$ coefficients, and a control that applies the offset to the axis command. Typically recovers 50–80% of the raw thermal error — the residual is the non-linear and gradient-driven part.

F-204

Bending caused by a temperature gradient

$$\frac{1}{R} = \frac{\alpha\, \Delta T_{grad}}{t} \;;\; \delta = \frac{L^2}{2R} \quad \text{[mm]}$$

ΔT_{grad} temperature difference across the member thickness t. A 2 K gradient through a 300 mm-deep, 1.5 m-long cast-iron beam bows it by 16 µm — larger than the uniform-growth effect of the same heat input. Sun on one side of a machine, or hot chips piled against one way cover, produces exactly this. Symmetric cooling and full enclosures are design requirements, not cosmetics.

WHERE THE HEAT COMES FROM

Heat source	Typical magnitude	Path into the structure	Design control
Spindle bearings	0.3 – 1.5 kW (F-111)	housing → headstock → column	oil-air or oil-mist lubrication, cooled oil jacket through the outer races, ceramic balls, preload chosen for the duty not the maximum
Ball screw friction	0.1 – 0.6 kW per axis (F-080)	nut and supports → bed	pre-tensioning to the expected rise (F-079), hollow screw with through-cooling, screw-temperature compensation in the control
Linear guide friction	0.05 – 0.3 kW per axis (F-194)	carriage → guideway → bed or column	way covers, symmetric cooling circuits on both sides of a column, low-preload grades where stiffness allows
Feed motor and drive losses	$(1-\eta)\cdot P_{in}$, 0.2 – 1 kW	motor flange → headstock or saddle	mount the motor off the accuracy-critical structure, forced air cooling, size for T_{rms} not peak so the motor runs cool
Hydraulic power unit	0.5 – 3 kW (F-209)	oil → reservoir → machine	variable-displacement pump with pressure compensation, oil cooler sized on total losses, reservoir and pump set outside the machine frame
Cutting process	15 – 30% of P_c (F-195)	chip → workpiece → coolant → tank	high-pressure through-tool coolant, chip conveyor that does not dump hot chips onto a way, full enclosure, coolant chiller
Control cabinet and drives	0.3 – 1.5 kW	cabinet → adjacent structure	cabinet heat exchanger, cabinet placed away from the column and the scales, filtered forced ventilation
Ambient and solar	±2 – 5 K	whole structure, asymmetrically	full enclosure, shop temperature held to ±1 K for precision machines, no direct sun or doorway draught on the machine

> **The thermal design rule.** Total heat input is easy to reduce; **asymmetry** is what actually destroys accuracy. A uniform 5 K rise in a well-designed machine moves the tool centre point predictably and can be compensated by F-203. A 2 K gradient across one column (F-204) bends it, and no compensation model built from a few temperature sensors will recover that reliably. Design for symmetry of heat input and of cooling first, and for compensation second.

SECTION 13

Hydraulics & pneumatics

F-205 – F-215
11 formulas

Clamping, counterbalancing, tool changing, tailstocks, chucks, pallet changers, way lubrication and brakes. Compact, high force density, and the source of most of a machine's heat if sized carelessly.

F-205

Cylinder force

$$F = p\,A\,\eta_m \;;\quad A_{bore} = \frac{\pi D^2}{4} \;;\quad A_{rod} = \frac{\pi(D^2 - d^2)}{4} \quad [\text{N}]$$

p working pressure (N/mm²; 1 bar = 0.1 N/mm²) · η_m mechanical efficiency 0.85–0.95 · D bore, d rod diameter. The annular (rod) side gives 20–40% less force — always check the retracting direction, which is usually the one doing the work in a clamp.

F-206

Cylinder speed and stroke time

$$v = \frac{Q}{A} \;;\quad t = \frac{A\,L}{Q} \;;\quad v\,[\text{m/min}] = \frac{60\,Q}{1000\,A} \quad (Q \text{ in L/min}, A \text{ in mm}^2)$$

L stroke (mm). Clamp cycle time is part of the non-cutting time in F-043 — a 3 s clamp on a 20 s cycle costs 15% of productivity.

F-207

Pump delivery

$$Q = V_g\,n\,\eta_v \quad [\text{L/min}]$$

V_g displacement per revolution (cm³/rev) · n pump speed (rpm) · η_v volumetric efficiency 0.90–0.97 (gear), 0.92–0.96 (vane), 0.93–0.98 (piston). A variable-displacement pump with pressure compensation idles at near-zero flow and is the right choice for a clamp-and-hold duty.

F-208

Hydraulic power

$$P = \frac{p\,Q}{600} \quad [\text{kW; } p \text{ in bar, } Q \text{ in L/min}]$$

A 40 L/min unit at 100 bar draws 6.7 kW, essentially all of which becomes heat unless the work absorbs it. This is why a fixed-displacement pump with a relief-valve bypass is thermally unacceptable on a modern machine — use variable displacement or an accumulator with an unloading valve.

F-209

Efficiencies of a hydraulic unit

$$\eta_t = \eta_v\,\eta_m \;;\quad \eta_m = \frac{\text{actual torque}}{\text{theoretical torque}} \;;\quad Q_{loss} = P_{in}(1 - \eta_t)$$

η_v volumetric (leakage), η_m mechanical/hydromechanical (friction). Total 0.80–0.90 for a good unit. Feed the losses into the oil-cooler sizing (F-199) — an undersized cooler is the most common cause of a machine that drifts all shift.

F-210

Pipe flow velocity and pressure drop

$$v = 21.2\,\frac{Q}{d^2} \quad [\text{m/s; } Q \text{ in L/min, } d \text{ in mm}] \;;\quad \Delta p = \lambda\,\frac{L}{d}\,\frac{\rho\,v^2}{2} \quad [\text{Pa}]$$

λ Darcy friction factor (0.02–0.04 turbulent, 64/Re laminar). Design velocities: suction line 0.5–1.5 m/s, pressure line 3–5 m/s, return line 1.5–3 m/s. Above these, pressure loss and — worse — cavitation and noise follow.

F-211

Reynolds number for hydraulic oil

$$\text{Re} = \frac{\rho\,v\,d}{\mu} = \frac{v\,d}{\nu} \;;\quad \text{laminar} < 2300 < \text{turbulent}$$

ν kinematic viscosity (m²/s); ISO VG32 oil at 40 C has $\nu \approx 32$ cSt $= 32 \times 10^{-6}$ m²/s. Cold start-up pushes a system laminar, changing pressure drops and valve response — a real cause of "the machine is slow for the first ten minutes".

F-212

Accumulator sizing (polytropic)

$$p_0 V_0^{\,n} = p_1 V_1^{\,n} = p_2 V_2^{\,n} \;;\quad \Delta V = V_1 - V_2 = V_0\,\left[\left(\frac{p_0}{p_2}\right)^{1/n} - \left(\frac{p_0}{p_1}\right)^{1/n}\right]$$

0 = precharge, 1 = minimum working pressure, 2 = maximum working pressure. $n = 1.0$ isothermal (slow duty, energy storage), $n = 1.4$ adiabatic (fast duty, shock absorption). Precharge is normally $0.9p_1$ for energy storage and $0.6–0.7p_2$ for shock damping. Used on a machine for clamp hold, counterbalance and emergency release.

F-213

Pneumatic cylinder force

$$F = p\,A\,\eta \quad \text{[N]}, \quad \eta \approx 0.85$$

p gauge pressure (N/mm²); at the usual 0.6 MPa a 50 mm bore gives ≈ 1.0 kN. Air is compressible, so a pneumatic clamp cannot hold a position against a varying cutting force — use it for actuation and detection, hydraulics or a mechanical toggle for clamping.

F-214

Free-air consumption of a pneumatic cylinder

$$Q_{\text{free}} = A\,L\,\frac{p + p_{\text{atm}}}{p_{\text{atm}}} = A\,L\,(p_{\text{rel}} + 1) \quad \text{[per stroke]}$$

p gauge, p_{atm} = 0.1013 MPa. Multiply by strokes per minute for the compressor duty. Air is roughly 10× the running cost of the equivalent electric actuation — a useful figure when justifying an electric tool changer or tailstock.

F-215

Pressure intensifier ratio

$$p_2 = p_1\,\frac{A_1}{A_2}\,\eta$$

Used to obtain a high clamping pressure (10–30 MPa) from a standard 7 MPa system for chucks, fixtures and drawbar boosters. Volume delivered falls in the same ratio as the pressure rises — check the cycle time with F-206.

SECTION 14

Precision, accuracy, tolerances & presses

F-216 – F-240
25 formulas

How the machine is measured and judged — ISO 230-2 positioning statistics, Abbe and cosine errors, tolerance grades and fits — followed by the force and energy relations for press-type machine tools.

POSITIONING ACCURACY (ISO 230-2)

F-216

Accuracy of positioning

$$A = E_{\max} - E_{\min}\ , \quad E = \max/\min \text{ over all targets of } (\bar{x}_i \pm 3S_i) \quad [\mu m]$$

$\bar{x}_i$ mean positional deviation at target i · S_i its standard deviation · both directions of approach included. A is the total band the axis occupies, not the largest single error — a machine can have small errors at every target and still fail A if the errors do not repeat.

F-217

Repeatability of positioning

$$R_i\uparrow = 5\,S_i\uparrow\ ; \quad R = \max_i\,(R_i\uparrow, R_i\downarrow, R_i^{B}) \quad [\mu m]$$

Five standard deviations covers 99.7% of a normal distribution with margin. Minimum five measurements per target, minimum five targets spread over the stroke. Repeatability is set by friction, backlash, scale resolution and structural compliance — and it is the parameter that determines whether a part can be made consistently, far more than accuracy does, because accuracy can be compensated and repeatability cannot.

F-218

Bidirectional repeatability and mean reversal value

$$R_i^{B} = \left|\bar{x}_i\uparrow - \bar{x}_i\downarrow\right| + 3S_i\uparrow + 3S_i\downarrow\ ; \quad B_i = \bar{x}_i\uparrow - \bar{x}_i\downarrow \quad [\mu m]$$

B_i is the reversal (backlash) value and is the direct measure of the lost motion that backlash compensation must remove. A large B with small S is a compensation problem; a large S is a friction, stiffness or thermal problem and cannot be compensated.

F-219

Standard deviation of the positional deviation

$$S_i = \sqrt{\frac{\Sigma_j\,(x_{ij} - \bar{x}_i)^2}{n-1}} \quad [\mu m]$$

x_{ij} the j-th of n measurements at target i. Report S_i per target, not pooled — a single bad target (usually where the screw support or a way joint sits) is the whole story.

GEOMETRIC ERROR SOURCES

F-220

Abbe error

$$\varepsilon_{\text{Abbe}} = L\sin\theta \approx L\,\theta\ ; \quad \varepsilon\ [\mu m] = L\ [mm] \times \theta\ [\mu rad] \times 10^{-3}$$

L offset between the measuring scale and the line of action of the cutting force · θ angular error of the moving element (pitch, yaw or roll). This is the dominant error on most machines: a 300 mm Abbe offset with a 10 µrad angular error gives 3 µm. **Design rule: put the scale in line with the tool whenever the layout allows** — Abbe-compliant design beats Abbe compensation.

F-221

Cosine error

$$\varepsilon_{\cos} = L(1-\cos\theta) \approx \frac{L\,\theta^2}{2}$$

Second-order in θ, therefore small: a 1 mrad misalignment of a 1 m laser interferometer gives only 0.5 µm. Unlike Abbe error it is always positive (the measurement reads short) and can be corrected by alignment rather than by compensation.

F-222

Thermal error of a measured length

$$\varepsilon_{\text{th}} = (\alpha_{\text{work}} - \alpha_{\text{machine}})\,L\,\Delta T + \alpha_{\text{machine}}\,L\,(T - 20\ °C)$$

Two distinct terms: differential expansion between workpiece and machine, and absolute departure of the machine from the 20 C reference. When machining aluminium (α = 23 × 10^{-6}/K) on a cast-iron machine (10.5 × 10^{-6}/K) a 5 K difference produces 63 µm/m of apparent size error — larger than the tolerance on most aluminium parts.

F-223

Feedback resolution

$$\text{res} = \frac{l}{4\,P\,n_{\text{int}}} \quad [mm]\ ; \quad \text{res}_{\text{linear scale}} = \frac{\text{grating pitch}}{n_{\text{int}}}$$

l screw lead (mm) · P encoder pulses per revolution · 4 = quadrature multiplication · n_{int} electronic interpolation factor (typically 4–4096). Rule of thumb: resolution should be one fifth to one tenth of the specified positioning accuracy, so a ±5 µm machine needs 0.5–1 µm resolution — 0.1 µm or better on a precision grinder.

F-224

Lost motion (backlash) of a drive train

$$L_{\text{lost}} = \Sigma\, \theta_{\text{back,i}} \frac{l}{2\pi\, i_i} + L_{\text{screw}} + L_{\text{guide}} \quad \text{[mm]}$$

$\theta_{\text{back,i}}$ angular backlash of each element (rad) · i_i its ratio to the screw · L_{screw} axial play of the nut and supports · L_{guide} clearance in the guideway. A preloaded double nut removes L_{screw}; a split pinion or a preloaded gear removes gear backlash; only a scale closed loop removes the residual, which is why full-closed-loop machines are specified for contouring accuracy.

TOLERANCES AND FITS

F-225

Standard tolerance unit (ISO 2892)

$$i = 0.45\, \sqrt[3]{D} + 0.001\, D \quad \text{[µm]} \;;\quad D = \sqrt{D_1\, D_2}$$

D is the geometric mean of the limits of the diameter step (e.g. for the 18–30 mm step, D = √(18×30) = 23.2 mm). The first term represents manufacturing difficulty, the second the measurement and thermal sensitivity of larger sizes.

F-226

IT tolerance grade

$$\text{IT}n = a_n\, i \;,\quad a\text{: IT5 = 7, IT6 = 10, IT7 = 16, IT8 = 25, IT9 = 40, IT10 = 64, IT11 = 100, IT12 = 160 ...}$$

Multipliers increase by a factor of ≈ 1.6 per grade. Machine-tool relevance: spindle bearing seats j5/k5 (IT5), gear bores H7, tool tapers to the taper standard, general machined features IT7–IT8, welded frames IT11–IT13 before machining. Specifying a tighter grade than the process can hold is pure cost.

F-227

Fit — clearance, interference and fit tolerance

$$X_{\min} = D_{\min} - d_{\max} \;;\quad N_{\max} = d_{\max} - D_{\min} \;;\quad T_{\text{fit}} = T_{\text{hole}} + T_{\text{shaft}}$$

Positive X = clearance, positive N = interference. The hole-basis system (H hole) is standard because reamers and broaches are fixed-size tools; the shaft is then varied to obtain the fit. The fit tolerance band is the sum of the two component tolerances — a fact often forgotten when a design "will not assemble".

F-228

Assembly temperature for a shrink or expansion fit

$$\Delta T = \frac{\delta + \delta_{\text{asm}}}{\alpha\, d} \quad \text{[K]}$$

δ required diametral interference · δ_{asm} assembly clearance, 0.05–0.15 mm · d interface diameter. Heating the hub: keep below 120 C for through-hardened bearing rings and below the tempering temperature of any hardened part; cooling the shaft in dry ice (−78 C) or liquid nitrogen (−196 C) is preferred for bearings because heating risks dimensional change and loss of hardness.

F-229

Error budget combination

$$\text{statistical (RSS): } \varepsilon_{\text{tot}} = \sqrt{\Sigma\, \varepsilon_i^2} \;;\quad \text{worst case: } \varepsilon_{\text{tot}} = \Sigma\, |\varepsilon_i|$$

Build the budget from: geometric errors of each axis (6 per axis: 3 linear, 3 angular), Abbe offsets, squareness between axes, thermal growth, drive compliance, backlash, scale error and fixture/workpiece deflection. Use RSS for independent random errors and worst-case summation for systematic ones (squareness, Abbe) — most machine budgets need both, applied term by term.

PRESSES AND FORMING MACHINES

F-230

Blanking and punching force

$$F = L\, t\, \tau_{\text{u}} = \pi\, d\, t\, \tau_{\text{u}} \quad \text{[N]}$$

L perimeter of the cut (mm) · t sheet thickness (mm) · τ_{u} ultimate shear strength ≈ $0.7\sigma_{\text{UTS}}$. Shear on the punch (single-shear 25–40% reduction, double-shear up to 60%) reduces peak force but not the total work — the flywheel and motor see the same energy.

F-231

Stripper force

$$F_{\text{s}} = (0.10 \ldots 0.20)\, F_{\text{blank}}$$

Add to the press force for the return stroke and to the clamp/spring sizing. A spring-loaded stripper must be pre-compressed enough to hold the strip flat before the punch enters.

F-232

V-die bending force

$$F = \frac{k\, L\, t^2\, \sigma_{\text{UTS}}}{W} \quad \text{[N]}$$

L bend length · t sheet thickness · W die opening width · $k \approx 1.33$ for air bending. Springback angle $\Delta\theta$ grows with σ_{UTS}, with W/t and falls with bend radius — the die designer overbends by 2–6° for mild steel and up to 15° for high-strength and aluminium alloys.

F-233

Press tonnage

$$\text{tonnage} = \frac{F}{9807} \quad \text{[metric tons; } F \text{ in N]}$$

1 tonne-force = 9.807 kN. Apply a 25–30% margin above the calculated peak force so the press works below its rated capacity at the point in the stroke where the force actually occurs — see F-240.

F-234

Deep-drawing force (ideal work balance)

$$F_{\text{draw}} = \pi\, d_{\text{p}}\, t\, \bar{\sigma}_{\text{u}} \ln \frac{D_{\text{b}}}{d_{\text{p}}} \quad \text{[N]}$$

d_{p} punch diameter · D_{b} blank diameter · $\bar{\sigma}_{\text{u}}$ mean flow stress over the drawing strain. Frictionless ideal; multiply by 1.5–2.0 for die-radius bending and blank-holder friction. Limiting drawing ratio $D_{\text{b}}/d_{\text{p}} \approx 2.0$ in one draw without a redraw.

F-235

Blank-holder force

$$F_{\text{BH}} = A_{\text{BH}}\, p_{\text{BH}} \quad \text{[N]}$$

A_{BH} area of the blank under the holder (mm²) · p_{BH} 2–3 N/mm² for mild steel, 3–4.5 for stainless, 1.5–2.5 for aluminium. Too little and the wall wrinkles; too much and the flange cannot draw in and the cup splits. A cushion with a controlled force–stroke curve is a press design parameter, not an accessory.

F-236

Flywheel stored energy and inertia

$$E = \frac{1}{2} I\, \omega^2 \quad \text{[J]} \;;\quad I_{\text{hollow}} = \frac{m(d_{\text{o}}{}^2 + d_{\text{i}}{}^2)}{8} \;;\quad I_{\text{disc}} = \frac{m\, d^2}{8}$$

ω in rad/s. Energy scales with the **square** of speed, so a smaller, faster flywheel stores the same energy — limited by the rim bursting stress $\sigma = \rho v^2_{\text{rim}}$, which caps cast-iron rims near 30–40 m/s and steel near 60–100 m/s.

F-237

Energy released by flywheel speed drop

$$E = I\, \omega_{\text{m}}{}^2\, C_{\text{s}}\,, \quad C_{\text{s}} = \frac{\omega_{\text{max}} - \omega_{\text{min}}}{\omega_{\text{m}}}$$

C_{s} coefficient of speed fluctuation: 0.10–0.20 for presses, 0.02 for machine-tool spindles (where speed constancy matters), 0.20–0.30 where it does not. Setting C_{s} is the flywheel sizing decision.

F-238

Press motor power

$$P = \frac{E}{t_{\text{cycle}}\, \eta} \quad \text{[W]}$$

E energy per stroke · t_{cycle} time between strokes. The motor supplies the **average** power; the flywheel supplies the peak. A press needing 500 kJ per stroke at 60 strokes/min needs only 167 kW of motor — provided the flywheel can be re-accelerated between strokes.

F-239

Ram kinetic energy at impact

$$E_{\text{ram}} = \frac{1}{2}\, m\, v^2 \quad \text{[J]}$$

For a hydraulic or servo press (no flywheel) this plus the stored pressure energy is what is available for forming. High-speed blanking relies on impact energy; deep drawing relies on sustained force at low speed — the two need different machines.

F-240

Available force over the working stroke

$$F_{\text{avail}} = \frac{E_{\text{avail}}}{s} \quad \text{[N]} \;;\quad \text{check } F_{\text{avail}}(s) \geq F_{\text{required}}(s) \text{ over the whole stroke}$$

s stroke distance over which the work is done. A press rated 1000 kN at 6 mm before bottom dead centre may deliver only a fraction of that higher up the stroke. Plot the machine's force–stroke curve against the process force–stroke curve — the classic and most-missed press selection check.

APPENDIX A

Worked example — sizing a machining-centre X axis

12 steps
21 formulas used

A complete feed-axis calculation for a vertical machining centre, in the order a designer would actually perform it. Every step cites the formula used. The intermediate results are carried forward, so the arithmetic can be checked line by line.

Design specification

Parameter	Value	Source
Axis stroke	800 mm	machine size
Moving mass (table + fixture + workpiece)	600 kg	layout estimate
Rapid traverse	36 m/min = 600 mm/s	specification
Maximum cutting feed	6 m/min = 100 mm/s	specification
Target acceleration	5 m/s² (0.5 g)	structural limit
Maximum cutting force in the axis direction	4000 N	Section 1, F-015
Guideway	2 × recirculating ball, 4 blocks, μ = 0.005	selected
Ball screw	40 mm nominal × 12 mm lead, root d_r = 33 mm	first choice
Screw mounting	fixed–supported, max. unsupported span 1000 mm	layout
Duty cycle	60 s: 2 rapids + 25 s cutting + idle	part programme
Life target	≥ 20 000 h	specification

STEP 1 **Axial force the drive must produce · F-068**

$F_a = F_{cut} + \mu W + ma = 4000 + (0.005 \times 600 \times 9.81) + (600 \times 5) = 4000 + 29 + 3000 =$ **7029 N**

The inertia term is 43% of the total. On this axis, acceleration — not cutting — sizes the drive.

STEP 2 **Ball screw drive torque · F-065**

$T_1 = F_a l/(2\pi\eta) = 7029 \times 0.012 / (2\pi \times 0.92) =$ **14.6 N·m** → with 8% bearing and seal friction: **15.8 N·m**

STEP 3 **Screw speed at rapid traverse**

$n = v_f/l = 600 / 12 = 50$ rev/s = **3000 rpm** (cutting feed: 100/12 × 60 = **500 rpm**)

STEP 4 **Critical speed check · F-069, F-070**

$n_{cr} = f\, d_r/L^2 \times 10^7 = 17.7 \times 33 / 1000^2 \times 10^7 = 5841$ rpm

$n_{perm} = 0.8 \times 5841 =$ **4673 rpm** > 3000 rpm ✓

$DN = 40 \times 3000 =$ **120 000** ≤ 150 000 (standard precision nut) ✓

Both speed limits are satisfied with margin. Had the stroke been 1500 mm, n_{cr} would fall to 2596 rpm and the rapid would have to be reduced or the screw diameter increased — the $1/L^2$ penalty of F-069.

STEP 5 **Motion profile · F-091, F-092**

$t_a = v/a = 0.6/5 =$ **0.12 s**; $s_a = v^2/(2a) = 0.36/10 =$ **36 mm**
$T_{rapid} = 800/600 + 0.6/5 = 1.333 + 0.120 =$ **1.45 s per traverse**

STEP 6 **Inertia referred to the motor shaft · F-086, F-087**

Element	Calculation	J (kg·m²)
Screw shaft (Ø40 × 1100, steel)	$m = 10.85$ kg; $J = md^2/8$	21.7×10^{-4}
Table, 600 kg on a 12 mm lead	$J = m(l/2\pi)^2 = 600 \times (0.00191)^2$	21.9×10^{-4}
Coupling and encoder	catalogue	4.0×10^{-4}
Total load inertia		$\mathbf{47.6 \times 10^{-4}}$

Note how closely the screw's own inertia matches the reflected table inertia — both must be counted.

STEP 7 **Acceleration and peak torque · F-084, F-085**

$T_a = J\Delta n/(9.55\ t_a) = 47.6\times10^{-4} \times 3000 / (9.55 \times 0.12) =$ **12.5 N·m**
$T_{peak} = T_a + T_{load} = 12.5 + 15.8 =$ **28.3 N·m**

STEP 8 **RMS torque over the duty cycle · F-085**

Segment	t (s)	T (N·m)	T^2t
Rapid acceleration (×2)	0.24	28.3	192
Rapid at constant speed (×2)	2.43	15.8	607
Rapid deceleration (×2)	0.24	12.5	38
Cutting feed, 6 m/min	25.00	9.0	2025
Idle, dwell, tool change	32.09	2.0	128
Total	**60.00**	–	**2990**

$T_{rms} = \sqrt{(2990/60)} =$ **7.1 N·m**

Peak torque is four times the RMS value — a typical machine-tool duty. The motor must satisfy both limits, and the drive's overload curve, not its nameplate, decides the peak.

STEP 9 **Motor selection and inertia ratio · F-089, F-105**

Required: rated ≥ 7.1 N·m, peak ≥ 28.3 N·m at 3000 rpm → a 1.2 kW AC servo (rated 8.0 N·m, peak 24–32 N·m, $J_m = 6.2 \times 10^{-4}$ kg·m²)
Inertia ratio = 47.6 / 6.2 = **7.7** ≤ 10 (high-dynamic class) ✓

STEP 10 **Mean load and rating life · F-075, F-073, F-074**

Cubic mean load over the cycle (rapid 3029 N @ 3000 rpm; cutting 4029 N @ 500 rpm; idle 300 N @ 50 rpm):
$F_m = \sqrt[3]{(\Sigma F^3 nt/\Sigma nt)} = \sqrt[3]{(1.060\times10^{15} / 22\,835)} =$ **3595 N**
Mean speed $n_m = \Sigma nt/\Sigma t = 22\,835/60 =$ **381 rpm**
With $C_a = 34\,000$ N: $L = (34\,000/3595)^3 \times 10^6 = 846 \times 10^6$ rev
$L_h = 10^6/(60 \times 381) \times 846 =$ **37 000 h** $\geq 20\,000$ h ✓

STEP 11 **Stiffness, positioning error and resonance · F-076, F-077, F-171**

Shaft (worst case, nut 700 mm from the fixed bearing): $K_s = AE/L = 855 \times 207\,000/700 =$ **253 N/µm**
Preloaded double nut $K_n = 400$ N/µm; fixed bearing pair $K_b = 500$ N/µm
$1/K = 1/253 + 1/400 + 1/500 \rightarrow K_{axis} =$ **118 N/µm**
Positioning error under 4000 N cutting force: $\varepsilon = 4000/118\,000 =$ **34 µm**
Axial resonance: $f_n = (1/2\pi)\sqrt{(K/m)} = (1/2\pi)\sqrt{(1.18\times10^8/600)} =$ **71 Hz**

Design decision. A 34 µm load-dependent error cannot be compensated by the screw encoder, because the encoder sits on the motor side of the compliance. Two options: (a) fit a linear scale on the table for full closed-loop control, or (b) pre-tension the screw (F-079) to double K_s, reducing the error to 26 µm, and still fit the scale. The 71 Hz axial resonance also caps the position-loop gain at roughly $K_v \approx 6$–$8\ s^{-1}$ (F-095), giving a following error of 100/8 = 12.5 µm at 6 m/min — acceptable for milling, not for jig boring.

STEP 12 **Buckling, thermal growth and guideway life · F-071, F-078, F-079, F-081**

Buckling (fixed–supported, $K = 0.7$): $I = \pi\times33^4/64 = 58\,213$ mm⁴
$P_{cr} = \pi^2 EI/(KL)^2 = \pi^2 \times 207\,000 \times 58\,213/(700)^2 =$ **243 kN**; $n_s = 243\,000/(7029 + 6200) =$ **18** ✓

Pre-tension for a 3 K expected rise: $F_{pre} = EA\alpha\Delta T = 207\,000 \times 855 \times 11.7\times10^{-6} \times 3 =$ **6200 N**
Uncompensated growth at $\Delta T = 4$ K over 800 mm: $\Delta L = 11.7\times10^{-6} \times 800 \times 4 =$ **37 µm** → pre-tensioning plus screw cooling is mandatory.

Guideway: worst block load 3500 N, $f_w = 1.5$, $C = 33$ kN → $L = 50(33\,000/(1.5 \times 3500))^3 =$ **12 400 km**; at 246 m/h of travel = **50 000 h** ✓

Result summary — X axis

Item	Result	Status
Ball screw	Ø40 × 12 lead, pre-tensioned 6.2 kN	accepted
Screw mounting	fixed–supported, 1000 mm span	accepted
Peak / RMS torque	28.3 / 7.1 N·m	accepted
Motor	1.2 kW AC servo, J_m = 6.2 × 10^{-4} kg·m²	accepted
Inertia ratio	7.7	accepted (≤ 10)
Screw life	37 000 h	accepted (≥ 20 000 h)
Guideway life	50 000 h	accepted
Buckling safety	18	accepted (≥ 4)
Axis stiffness	118 N/µm	**marginal**
Load-dependent position error	34 µm	**action: linear scale required**
Axial resonance	71 Hz	**limits K_v to ≈ 8 s^{-1}**
Thermal growth, uncompensated	37 µm at +4 K	**action: pre-tension + cooling**

> **What the example shows.** Nine of the thirteen checks passed on the first pass. The four that did not are all in the same family — stiffness, resonance and thermal growth — and none of them is visible in a torque-and-life calculation alone. On a real project these are the items that decide whether the machine meets its accuracy specification, and they must be evaluated at the concept stage, when the screw diameter, the bearing span and the scale position can still be changed.

APPENDIX B

Design data tables

9 tables
constants & properties

Representative values for the constants that appear in the formulas. They are starting points for a first calculation, not substitutes for the measured data of the specific material, tool grade or component you have selected — confirm every critical value against the supplier's current catalogue or a test on the actual material.

B1 Specific cutting force $k_{c1.1}$ (at 1.0 mm chip thickness) and thickness exponent m_c — for F-008, F-009, F-010

Workpiece material	Typical UTS (N/mm²)	$k_{c1.1}$ (N/mm²)	m_c	Notes
Low-carbon steel S235 / 1018	370–440	1900–2200	0.21	free cutting, long chip
Medium-carbon steel C45 / 1045	570–700	2400–2700	0.21	—
Alloy steel 42CrMo4 / 4140	800–1000	2800–3200	0.22	annealed condition
Quenched & tempered 900 N/mm²	900–1100	3000–3400	0.23	—
Austenitic stainless 1.4301 / 304	550–750	2600–2900	0.18	work-hardens; high F_p
Duplex stainless 1.4462	700–900	3100–3500	0.18	low speed, rigid set-up
Grey cast iron EN-GJL-250	250 (bend)	1300–1500	0.19	segmented chip, abrasive
Ductile iron EN-GJS-500	500–700	1800–2100	0.20	—
Wrought aluminium AlMgSi	240–310	650–800	0.15	very high speed possible
Cast Al–Si (with Si > 10%)	180–280	800–1200	0.15	abrasive; PCD only
Brass CuZn39Pb3	380–500	700–900	0.15	short chip
Tin bronze CuSn12	300–400	1100–1400	0.17	—
Titanium Ti-6Al-4V	900–1100	2100–2600	0.20	low k, but poor conductivity → heat in the tool
Nickel alloy Inconel 718	1000–1300	3300–4000	0.25	low speed, high rigidity essential
Hardened steel 55 HRC	≈ 1900	3000–3400	0.25	hard turning; CBN

B2 Taylor tool-life exponent n (F-034 … F-036) and wear criterion

Tool / workpiece combination	n	Usual wear criterion
HSS on steel	0.10 – 0.15	VB = 0.3–0.6 mm
Uncoated carbide on steel	0.20 – 0.30	VB = 0.3 mm
Coated carbide (TiAlN) on steel	0.25 – 0.35	VB = 0.3 mm
Coated carbide on cast iron	0.20 – 0.30	VB = 0.3 mm or flank cracking
Ceramic on hardened steel / cast iron	0.40 – 0.60	notching, comb cracking
CBN on hardened steel	0.25 – 0.40	VB = 0.15–0.2 mm
PCD on aluminium	0.20 – 0.30	VB = 0.2 mm
Carbide / CBN on titanium	0.15 – 0.25	diffusion wear, VB = 0.2 mm
Carbide / ceramic on Inconel	0.15 – 0.25	notch wear at depth-of-cut line
Carbide end mill, slotting (full engagement)	0.18 – 0.25	VB = 0.25 mm

B3 Preferred spindle speed series (R20, $\varphi = 1.12$) — for F-052, F-055

rpm	rpm	rpm	rpm	rpm	rpm	rpm
45	50	56	63	71	80	90
100	112	125	140	160	180	200
224	250	280	315	355	400	450
500	560	630	710	800	900	1000
1120	1250	1400	1600	1800	2000	2240
2500	2800	3150	3550	4000	4500	5000
5600	6300	7100	8000	9000	10000	11200
12500	14000	16000	18000	20000	22400	25000

B4 Properties of machine-structure materials — for F-135 … F-142, F-166, F-196, F-201

Material	E (GPa)	G (GPa)	ρ (kg/m³)	ν	α (10^{-6}/K)	k (W/m·K)	ψ_{damp}
Grey cast iron EN-GJL-250	100–140	40–54	7100–7300	0.21–0.27	9.0–10.5	48–55	0.03–0.10
Ductile iron EN-GJS-500	165–180	65–70	7100	0.28	11–12	36	0.02–0.05
Structural steel S355 (welded frame)	210	81	7850	0.30	11.7	50	0.002–0.01
Alloy steel 42CrMo4 (qt)	210	81	7850	0.29	11.5	42	0.002
Bearing steel 100Cr6 / 52100	207	80	7810	0.29	11.9	46	0.002
Aluminium alloy 6082-T6	70	26	2700	0.33	23.5	170	0.001
Mineral casting / epoxy granite	15–45	6–18	2200–2400	0.20	10–18	1.0–1.5	0.05–0.15
Natural granite (metrology frames)	50–70	22–28	2650	0.20	6–8	2.5	0.005
Silicon nitride Si_3N_4 (ceramic balls)	310	125	3200	0.26	3.2	20	–

ψ_{damp} = specific damping capacity (F-139). Mineral casting trades a low E for three to ten times the damping of cast iron and a third of the density — the usual choice for high-speed milling and grinding frames where dynamic behaviour, not static stiffness, governs.

B5 Coefficients of friction and efficiencies — for F-068, F-085, F-125, F-157, F-185

Contact or element	μ static	μ kinetic / η
Steel on steel, dry	0.50 – 0.80	0.40 – 0.60
Steel on steel, oiled	0.10 – 0.15	0.06 – 0.10
Cast iron on hardened steel way, lubricated	0.15 – 0.25	0.08 – 0.15
PTFE-lined composite on hardened steel way	0.08 – 0.12	0.05 – 0.08
Recirculating ball linear guide	–	0.002 – 0.005
Recirculating roller linear guide	–	0.002 – 0.004
Hydrostatic way or bearing	0 (no static friction)	0.0005 – 0.001
Aerostatic way or bearing	0	≈ 0.0001
Ball screw (recirculating)	–	η = 0.90 – 0.95
Trapezoidal / Acme sliding screw, oiled	–	η = 0.30 – 0.50
Precision angular-contact ball bearing	–	μ = 0.0010 – 0.0015
Cylindrical roller bearing	–	μ = 0.0015 – 0.0025
Thrust bearing (drawbar)	–	μ = 0.002 – 0.004
Spur / helical gear mesh, oil lubricated	–	μ = 0.05 – 0.10; η = 0.985 – 0.995
V-belt on cast-iron pulley	0.30 – 0.40	η = 0.95 – 0.98
Toothed (synchronous) belt	–	η = 0.96 – 0.98
Roller chain	–	η = 0.96 – 0.98
7:24 taper tool connection (dry)	0.08 – 0.15	lower when lubricated
Bolt nut factor K — dry zinc plated	–	0.18 – 0.22
Bolt nut factor K — lightly oiled	–	0.14 – 0.16
Bolt nut factor K — MoS_2 paste	–	0.10 – 0.12
Press fit, steel on steel (assembly)	0.10 – 0.14	0.08 – 0.10 after shrink

B6 Guideway types compared — for F-185 … F-194

Type	Friction μ	Stiffness (N/µm)	Damping	Max speed	Typical application
Sliding, cast iron on steel	0.08–0.15	50–200	high	15 m/min	heavy-duty lathes, planers, stick-slip risk
Sliding, PTFE-lined	0.05–0.08	40–150	high	30 m/min	general machining centres, low cost
Rolling, ball	0.002–0.005	30–120	low	120 m/min	high-speed milling, standard machining centres
Rolling, roller	0.002–0.004	100–400	low	100 m/min	heavy milling, high stiffness with low friction
Hydrostatic	≈ 0.0005	200–1000	very high	60 m/min	precision grinders, heavy boring mills, jig borers
Aerostatic	≈ 0.0001	5–40	low	100 m/min	ultra-precision, diamond turning, CMMs, light loads only

B7 End-fixity constants — for F-069, F-071, F-180

Support condition	Effective length factor K	$(\lambda L)^2$	f in F-069	Ball screw mounting
Fixed – free (cantilever)	2.00	3.52	3.9	unsupported end
Supported – supported (pinned)	1.00	9.87	11.5	both ends on simple bearings
Fixed – supported	0.70	15.4	17.7	**standard machine-tool arrangement**
Fixed – fixed	0.50	22.4	25.5	pre-tensioned, both ends axially locked
Fixed – pinned (buckling)	0.70	–	–	—

B8 ISO 286 standard tolerances (μm) for selected size ranges — for F-225, F-226

Size range (mm)	IT5	IT6	IT7	IT8	IT9	IT10	IT11
over 0 to 3	4	6	10	14	25	40	60
over 3 to 6	5	8	12	18	30	48	75
over 6 to 10	6	9	15	22	36	58	90
over 10 to 18	8	11	18	27	43	70	110
over 18 to 30	9	13	21	33	52	84	130
over 30 to 50	11	16	25	39	62	100	160
over 50 to 80	13	19	30	46	74	120	190
over 80 to 120	15	22	35	54	87	140	220
over 120 to 180	18	25	40	63	100	160	250
over 180 to 250	20	29	46	72	115	185	290
over 250 to 315	23	32	52	81	130	210	320
over 315 to 400	25	36	57	89	140	230	360

B9 Unit conversions used in this handbook

Quantity	Conversion
Pressure	1 bar = 0.1 N/mm² = 10^5 Pa = 14.50 psi; 1 MPa = 1 N/mm² = 10 bar
Force	1 kgf = 9.807 N; 1 tonne-force (tf) = 9807 N; 1 lbf = 4.448 N
Torque	1 N·m = 0.7376 lbf·ft = 8.851 lbf·in; 1 kgf·m = 9.807 N·m
Power	1 kW = 1.360 PS = 1.341 hp; 1 kcal/h = 1.163 W
Speed	1 m/min = 0.0167 m/s; 1 m/s = 60 m/min; 1 rpm = 0.1047 rad/s
Flow	1 L/min = 1.667×10^{-5} m³/s; 1 gal(US)/min = 3.785 L/min
Viscosity	1 cSt = 1 mm²/s = 10^{-6} m²/s; μ (N·s/mm²) = ν (mm²/s) × ρ (kg/mm³)
Temperature	Δ1 K = Δ1 °C = Δ1.8 °F; reference temperature for dimensional metrology = 20 °C (ISO 1)
Angle	1 rad = 57.296°; 1° = 17.45 mrad; 1 μrad over 1 m = 1 μm (Abbe error)
Surface finish	$R_a \approx R_t/4$ for a regular feed-mark profile; 1 μm = 39.4 μin

APPENDIX C

Index of formulas

F-001 – F-240
by number

All 240 formulas in numerical order with the section in which they appear.

F-001 Cutting speed (peripheral speed of the cut)

F-002 Spindle speed required for a given cutting speed

F-003 Feed rate from feed per revolution — turning, boring, drilling

F-004 Feed rate from feed per tooth — milling

F-005 Material removal rate — turning

F-006 Material removal rate — milling

F-007 Material removal rate — drilling

F-008 Specific cutting force (unit cutting force)

F-009 Main cutting force — turning

F-010 Correction of kc for chip thickness

F-011 Cutting power

F-012 Motor power required

F-013 Torque from power and speed

F-014 Spindle torque from the cutting force

F-015 Resultant cutting force

F-016 Chip thickness ratio

F-017 Shear angle from the chip ratio

F-018 Merchant's minimum-energy relation

F-019 Shear strain in the primary deformation zone

F-020 Shear and normal force on the shear plane

F-021 Friction angle and coefficient at the rake face

F-022 Cutter engagement (contact) angle

F-023 Average chip thickness — milling

F-024 Effective number of teeth in cut

F-025 Approach distance — face milling

F-026 Drill point approach length

F-027 Drilling torque and thrust (empirical)

F-028 Drilling power

F-029 Grinding — wheel speed and speed ratio

F-030 Grinding — specific removal rate and specific energy

F-031 Grinding ratio (G-ratio)

F-032 Thread-cutting gear train ratio

F-033 Power-screw torque — taps, lead screws, clamping screws

F-034 Taylor's tool life equation

F-035 Extended tool life equation

F-036 Tool life at a changed cutting speed

F-037 Theoretical peak roughness — nose radius

F-038 Theoretical peak roughness — sharp tool with approach angles

F-039 Vibration-induced roughness contribution

F-040 Basic cutting time — turning

F-041 Basic cutting time — milling

F-042 Basic cutting time — drilling

F-043 Total cycle time per piece

F-044 Tool cost per piece

F-045 Cost per piece

F-046 Economic tool life (minimum cost per piece)

F-047 Tool life for maximum production rate

F-048 Optimum cutting speed

F-049 Overall equipment effectiveness

F-050 Speed range of the spindle

F-051 Geometric progression (structural) ratio

F-052 Value of the i-th speed step

F-053 Number of speed steps required

F-054 Loss of economic cutting speed

F-055 Preferred-number (Renard) series derivation

F-056 Transmission ratio of a gear pair

F-057 Overall ratio of a gear train

F-058 Structural formula and group transmission range

F-059 Practical ratio limits in a stepped gearbox

F-060 Centre distance and module

F-061 Constant sum of teeth within a group

F-062 Minimum number of teeth (no profile shift)

F-063 Feed range and feed progression ratio

F-064 Pitch diameter and gear peripheral speed

F-065 Ball screw drive torque

F-066 Back-drive (holding) torque

F-067 Screw efficiency from the lead angle

F-068 Axial force the drive must produce

F-069 Critical (whirling) speed of the screw shaft

F-070 Permissible rotational speed and DN limit

F-071 Buckling load of the screw shaft

F-072 Safety factor against buckling

F-073 Ball screw rating life

F-074 Rating life in operating hours

F-075 Mean (equivalent) axial load over a duty cycle

F-076 Axial stiffness of the screw system

F-077 Positioning error caused by drive compliance

F-078 Thermal elongation of the screw

F-079 Pre-tension force for thermal compensation

F-080 Heat generated by screw friction

F-081 Linear guideway rating life

F-082 Static safety factor of a guideway block

F-083 Equivalent block load with an overturning moment

F-084 Acceleration torque

F-085 Total and RMS motor torque

F-086 Inertia of a solid cylinder (shaft, screw, coupling)

F-087 Inertia of a translating mass referred to the screw

F-088 Inertia and torque through a reduction

F-089 Inertia ratio check

F-090 Rack-and-pinion and belt drive equivalence

F-091 Trapezoidal profile — acceleration time and distance

F-092 Total positioning time (trapezoidal, no jerk limit)

F-093 Acceleration achievable within the torque limit

F-094 Jerk-limited (S-curve) profile

F-095 Following (servo lag) error

F-096 Circularity error on a circular interpolation

F-097 Resultant feed rate of simultaneous axes

F-098 Block execution time

F-099 Circular interpolation — centre offsets and radius

F-100 Arc length and sweep time

F-101 Helical interpolation

F-102 Cutter radius compensation offset path

F-103 Polar / cylindrical coordinate transform

F-104 Regenerative braking energy and resistor rating

F-105 Motor torque–speed characteristic (two regions)

F-106 Basic rating life of a rolling bearing

F-107 Rating life in operating hours

F-108 Equivalent dynamic bearing load

F-109 Bearing speed limit — DN value

F-110 Centrifugal force on a bearing ball

F-111 Bearing friction torque and heat generation

F-112 Stiffness at the tool tip — rigid shaft on elastic bearings

F-113 Tool-tip deflection — elastic shaft, rigid bearings

F-114 Spindle thermal growth

F-115 Torque capacity of a taper tool connection

F-116 Required drawbar clamping force

F-117 Power delivered at the tool

F-118 Spindle acceleration time

F-119 Gear tooth bending stress (Lewis equation)

F-120 Dynamic (velocity) factor — AGMA

F-121 Gear contact (pitting) stress — ISO 6336 form

F-122 Involute gear geometry

F-123 Contact ratio

F-124 Gear mesh efficiency

F-125 Belt tension ratio (Euler–Eytelwein)

F-126 Belt power and centrifugal tension

F-127 Belt length over two pulleys

F-128 Chain speed, tension and polygonal fluctuation

F-129 Static stiffness

F-130 Combination of stiffnesses

F-131 Machine loop (tool–workpiece) stiffness

F-132 Dimensional error produced by compliance

F-133 Beam deflections — standard cases

F-134 End rotation of a cantilever

F-135 Torsional deformation

F-136 Torsional stiffness

F-137 Second moment of area — common machine sections

F-138 Joint (contact) stiffness

F-139 Damping capacity and damping ratio

F-140 Stiffness required by the accuracy specification

F-141 Stiffness scaling with wall thickness

F-142 Machine mass and foundation pressure

F-143 Direct stress, strain and Hooke's law

F-144 Elastic constant relation

F-145 Axial elongation

F-146 Bending stress and section moduli

F-147 Torsional shear stress

F-148 Von Mises equivalent stress

F-149 Principal stresses and maximum shear stress

F-150 Factor of safety

F-151 Stress concentration and notch sensitivity

F-152 Slenderness ratio and column failure mode

F-153 Fatigue criteria for combined mean and alternating stress

F-154 Modified endurance limit

F-155 Cumulative fatigue damage (Miner's rule)

F-156 Bolt preload

F-157 Bolt tightening torque

F-158 Bolt tensile stress area

F-159 Joint load factor and bolt load under external force

F-160 Key strength — shear and crushing

F-161 Shaft diameter for combined bending and torsion (ASME code form)

F-162 Combined critical speed of a multi-mass shaft (Dunkerley)

F-163 Fillet-weld throat stress

F-164 Press-fit contact pressure (Lamé)

F-165 Press-fit torque and axial capacity

F-166 Thermal stress in a constrained member

F-167 Hertzian contact — sphere on surface (point contact)

F-168 Hertzian contact — cylinders (line contact)

F-169 Helical compression spring

F-170 Bearing pressure and pv limit on sliding surfaces

F-171 Natural frequency of a single-degree-of-freedom system

F-172 Damping ratio from the logarithmic decrement

F-173 Quality factor and resonant amplitude

F-174 Dynamic magnification factor

F-175 Vibration transmissibility

F-176 Isolation efficiency

F-177 Mount natural frequency from static deflection

F-178 Torsional natural frequency

F-179 Two-mass torsional system

F-180 Shaft critical speed from static deflection (Rayleigh)

F-181 Regenerative chatter — limiting depth of cut

F-182 Chatter lobe — spindle speeds at a given chatter frequency

F-183 Tooth-passing frequency and force pulsation

F-184 Impact and shock load factor

F-185 Friction and total drive force

F-186 Way bearing pressure

F-187 Maximum edge pressure with an overturning moment

F-188 Stick-slip avoidance criterion

F-189 Hydrostatic pad load capacity

F-190 Oil flow through a bearing land (parallel plates)

F-191 Hydrostatic pad stiffness

F-192 Hydrostatic pump power and oil temperature rise

F-193 Aerostatic pad load, stiffness and air consumption

F-194 Rolling guideway friction power and heat

F-195 Heat generated by cutting and its partition

F-196 Conduction and thermal resistance

F-197 Convection

F-198 Steady-state temperature rise of a component

F-199 Coolant flow required for heat removal

F-200 Coolant temperature rise through the cut

F-201 Thermal growth of a machine element

F-202 Thermal time constant and warm-up curve

F-203 Thermal error compensation

F-204 Bending caused by a temperature gradient

F-205 Cylinder force

F-206 Cylinder speed and stroke time

F-207 Pump delivery

F-208 Hydraulic power

F-209 Efficiencies of a hydraulic unit

F-210 Pipe flow velocity and pressure drop

F-211 Reynolds number for hydraulic oil

F-212 Accumulator sizing (polytropic)

F-213 Pneumatic cylinder force

F-214 Free-air consumption of a pneumatic cylinder

F-215 Pressure intensifier ratio

F-216 Accuracy of positioning

F-217 Repeatability of positioning

F-218 Bidirectional repeatability and mean reversal value

F-219 Standard deviation of the positional deviation

F-220 Abbe error

F-221 Cosine error

F-222 Thermal error of a measured length

F-223 Feedback resolution

F-224 Lost motion (backlash) of a drive train

F-225 Standard tolerance unit (ISO 2892)

F-226 IT tolerance grade

F-227 Fit — clearance, interference and fit tolerance

F-228 Assembly temperature for a shrink or expansion fit

F-229 Error budget combination

F-230 Blanking and punching force

F-231 Stripper force

F-232 V-die bending force

F-233 Press tonnage

F-234 Deep-drawing force (ideal work balance)

F-235 Blank-holder force

F-236 Flywheel stored energy and inertia

F-237 Energy released by flywheel speed drop

F-238 Press motor power

F-239 Ram kinetic energy at impact

F-240 Available force over the working stroke

Design release checklist

Run this before releasing a feed axis, a spindle or a structure. Every line has a formula behind it.

Feed axis

✓	Check	Formula
☐	Axial force assembled from cutting, friction, inertia and gravity terms	F-068
☐	Peak and RMS motor torque both inside the drive's curves	F-084, F-085, F-105
☐	Inertia ratio within the drive manufacturer's limit	F-086 … F-089
☐	Critical speed and *DN* limit both clear of maximum screw speed	F-069, F-070
☐	Buckling safety factor ≥ 2 (≥ 4 for vertical axes)	F-071, F-072
☐	Screw and guideway life ≥ the specified hours on the real duty cycle	F-073 … F-075, F-081
☐	Axis stiffness computed at the **worst** table position, not mid-stroke only	F-076, F-077
☐	Thermal growth quantified; pre-tension, cooling or compensation decided	F-078 … F-080
☐	Axial resonance frequency at least 4× the intended position-loop bandwidth	F-171, F-178, F-179
☐	Regenerative energy absorbed by the drive or an external resistor	F-104
☐	Vertical axis: back-drive torque covered by brake, counterbalance or self-locking screw	F-066, F-067
☐	Crash load case checked against static guide rating	F-082, F-184

Spindle

✓	Check	Formula
☐	Power and torque quoted at the spindle nose across the whole speed range, not at the motor	F-011 … F-014, F-105, F-117
☐	Constant-torque / constant-power base speed matched to the intended cutting speeds	F-105
☐	Bearing life ≥ target at the duty-cycle mean speed and load	F-106 … F-108
☐	*DN* value and ball centrifugal force inside the bearing's limits	F-109, F-110
☐	Bearing friction heat computed and a cooling path provided	F-111, F-195 … F-198
☐	Tip stiffness computed with the real tool overhang, not a nominal one	F-112, F-113
☐	Thermal growth of the nose quantified and compensated or cooled	F-114, F-201
☐	Taper torque capacity above the maximum cutting torque with margin	F-115, F-116
☐	Ramp time verified with the chuck and maximum workpiece inertia included	F-118
☐	Stability lobe diagram produced for the intended tooling and material	F-181, F-182

Structure and accuracy

✓	Check	Formula
☐	Required stiffness derived from the accuracy budget **before** geometry is fixed	F-129, F-140
☐	Full load path modelled in series; softest element identified and addressed	F-130, F-131
☐	Bolted and slideway joints included in the stiffness model, not assumed rigid	F-138, F-159
☐	First three natural frequencies clear of tooth-passing frequencies over the whole speed range	F-171, F-174, F-183
☐	Damping strategy chosen (material, joints, fillers) and its effect quantified	F-139, F-172
☐	Abbe offsets minimised by layout; residual offsets listed in the error budget	F-220, F-229
☐	Thermal error budget built along the whole structural loop with symmetric cooling	F-201 … F-204
☐	Scale resolution ≤ 1/5 of the specified positioning accuracy	F-223
☐	Lost motion budgeted and a compensation or full-closed-loop strategy chosen	F-224, F-218
☐	Axis gains matched across all axes before the circularity test	F-095, F-096
☐	Floor loading and levelling-pad pressure inside the site limit	F-142
☐	ISO 230-2 acceptance test defined in the purchase specification, with the target values	F-216 … F-219

End of handbook. 240 numbered formulas across 14 sections, two worked design examples, nine data tables and a three-part release checklist. Formula numbers are stable and may be cited directly in calculation sheets, design reviews and supplier correspondence.